
On Variance Reduction in Learning Mean Flows

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Abstract

One-step generative modeling has emerged as a leading approach to amortize the inference cost of diffusion and flow-matching models. Among distillation-free methods, MeanFlow training is notoriously unstable, with non-decreasing loss and unbounded gradient variance. In this work, we establish a theory that attributes this pathology to a misuse of the conditional velocity field: it plays two distinct statistical roles in the loss, both as an unbiased regression target and as a Monte Carlo control variate inside a Jacobi-vector product, with the original loss assigning the wrong coefficient to the latter. We derive the optimal coefficient in closed form, and show that a family of fixes in concurrent works corresponds to different practical realizations of the same optimum. A controlled sweep of this coefficient on two-dimensional benchmarks and on a latent Diffusion Transformer recovers the predicted bias-variance ordering. The optimal coefficient yields up to a 54% improvement in sample quality on two-dimensional benchmarks and a monotone FID trend at every matched-step DiT checkpoint. Crucially, the same DiT measurement also reveals a quantitative *FID-MSE landscape mismatch*: although gradient variance is minimized at an interior coefficient value, the coefficient that minimizes FID prefers the direct use of conditional velocity.

1 Introduction

Deep generative models that leverage neural networks to parametrize and sample from unknown data distributions have achieved remarkable success [1, 2, 3, 4, 5]. Among these models, flow-based generative models [6, 7, 8, 9, 10, 11, 12] learn a continuous-time transformation between a simple prior and the data distribution. Conditional flow matching [10, 11] and stochastic interpolants [12, 13] further enable simulation-free training by regressing a velocity field, achieving sample quality competitive with diffusion models [1, 14, 2]. However, integration at inference time remains a challenge due to its computational costs and numerical instability, motivating works on *one-step generation*, including progressive distillation [15], distribution matching [16, 17], consistency models [18, 19], and flow map matching [20]. Nevertheless, most of them require a pre-trained distillation teacher.

Mean Flows [21] provide a *distillation-free* alternative by learning a two-parameter average velocity field $\mathbf{u}(\mathbf{x}, r, t)$ through regressing a total-derivative identity between the average and instantaneous velocity. In principle, this identity yields exact self-supervision. In practice, however, training is plagued by *non-decreasing loss* and prohibitively *high gradient variance* [22, 23]. A plethora of concurrent works have proposed remedies that target different mechanisms. AlphaFlow [22] attributes slow convergence to gradient conflict between flow-matching and total-derivative consistency components and proposes an α -curriculum. Improved MeanFlow [23] replaces the conditional-velocity tangent with a learned marginal-velocity head. Re-MeanFlow [24] preprocesses the data with rectified-flow straightening to reduce trajectory curvature. Terminal Velocity Matching [25] sidesteps the spatial Jacobian by differentiating with respect to the terminal time. Functional Mean Flows [26] extend MeanFlow to Hilbert spaces and study marginal-

conditional consistency for functional data. Each fix is empirically effective in isolation, but there is no theory that explains *why* the original objective is unstable or what these fixes have in common. In this paper, we establish our theory around a single research question:

Why do stochastic tangents destabilize learning Mean Flows?

We answer this question through the lens of *variance reduction*. Specifically, we identify that the conditional velocity $\mathbf{v}_{\text{cond}} = \mathbf{x}_1 - \mathbf{x}_0$ plays two statistically distinct roles in the MeanFlow loss. As the regression target, it is an *unbiased* estimator of the marginal velocity $\mathbf{v}$. As the tangent inside the Jacobian-vector product (JVP) of the total-derivative identity, however, the same $\mathbf{v}_{\text{cond}}$ acts as a Monte Carlo *control variate* for the inaccessible marginal velocity, with a statistically suboptimal coefficient. The bias-variance trade-off generated by this substitution governs gradient variance through the quadratic form $\mathbf{g}^\top \mathbf{J} \Sigma_{\mathbf{v}} \mathbf{J}^\top \mathbf{g}$, where $\mathbf{g} = \nabla_{\boldsymbol{\theta}} \mathbf{u}_{\boldsymbol{\theta}}$ and $\mathbf{J}$ is a spatial-Jacobi factor. The conditional fluctuation $\mathbf{v}_{\text{cond}} - \mathbf{v}$ is amplified by $\mathbf{J}$ at each gradient step, and stop-gradient prevents this amplification from reaching the optimizer, leading to the empirical pathology.

Following the theory, we derive the optimal tangent control-variate coefficient in closed form and show that it converges to a deterministic-tangent estimator in the practical regime where a marginal-velocity estimator is available. This result *explains* why deterministic tangents (*e.g.*, *EMA teachers and learned velocity heads*) all stabilize training: they are different practical instantiations of the same statistical optimum. To validate the framework empirically, we instantiate the $\beta = 1$ corner of β^* via an EMA-derived proxy with a flow-matching anchor (one extra JVP per step relative to vanilla MeanFlow), and run a controlled β -sweep ($\beta \in \{0, 0.25, 0.5, 1\}$) at toy and DiT-B/4 / ImageNet-256 scale. Direct measurement of per-step gradient variance recovers the predicted non-decreasing loss component $\text{Tr}(\mathbf{J} \Sigma_{\mathbf{v}} \mathbf{J}^\top)$ and a $1.7 \times \sim 4.7 \times$ reduction at $\beta = 1$ on a 2-D variance-probing sub-sweep. At DiT scale the same sweep also exposes a quantitative *FID-vs-MSE landscape mismatch*: the FID ordering matches the bias-variance prediction across all four β values, but the FID-axis offset ratio across $\beta = 0.5$ vs $\beta = 1$ is super-linear in the MSE-axis bias coefficient, so the FID-optimal β shifts past the gradient-MSE interior minimizer to the unbiased corner. Our contributions in this paper are as follows:

- C1** We prove that per-step gradient variance in the original MeanFlow scales unboundedly with $\mathbf{g}^\top \mathbf{J} \Sigma_{\mathbf{v}} \mathbf{J}^\top \mathbf{g}$ and the stop-gradient operator induces a *semi-gradient gap* that hides this term from the optimizer.
- C2** We frame the JVP tangent as a control variate for the marginal velocity, derive the optimal coefficient β^* in closed form, and unify concurrent works by showing that they each correspond to a different practical realization of β^* .
- C3** We empirically validate the framework via a controlled β -sweep at both 2-D and DiT-B/4 / ImageNet-256 scale, instantiating the $\beta = 1$ corner with an EMA proxy and flow-matching anchor. We observe $1.7 \times \sim 4.7 \times$ direct gradient-variance reduction, up to 54% SW_1 improvements on toy benchmarks, the predicted bias-variance FID ordering at every matched-step DiT checkpoint after the early-training transient, and a quantitative FID-vs-MSE landscape mismatch: the gradient-MSE optimum sits near $\beta \approx 0.94$ (directly measured matrix-form; scalar-isotropic prediction $\beta^* \in [0.46, 0.50]$) while the FID optimum lies at the unbiased corner $\beta = 0$, with a $\sim 6:1$ FID-offset ratio across $\beta = 0.5/\beta = 1$ versus the 1:4 MSE-axis prediction.

2 Preliminaries

Flow-Matching. Let $\mathbf{x}$ denote a data sample in $\mathcal{X} \subset \mathbb{R}^d$. Flow-matching generative models [10, 11] learn a time-dependent diffeomorphism $\psi(\mathbf{x}, t) : \mathbb{R}^d \times [0, 1] \rightarrow \mathbb{R}^d$ from a simple distribution $\mathbf{x}_1 \sim p(\mathbf{x}, 1)$ to the target $\mathbf{x}_0 \sim p(\mathbf{x}, 0) = p_{\text{data}}$ by parameterizing a velocity field $\mathbf{v}(\mathbf{x}, t) \triangleq \frac{d}{dt} \psi(\mathbf{x}, t)$. Since the exact *marginal velocity field* $\mathbf{v}(\mathbf{x}, t)$ is generally inaccessible, conditional flow matching instead learns a conditional field $\mathbf{v}(\mathbf{x}, t \mid \mathbf{x}_0) = \mathbf{x}_1 - \mathbf{x}_0$ with $\mathbf{x}_t = (1 - t)\mathbf{x}_0 + t\mathbf{x}_1$. The following lemma justifies the substitution. See Appendix C for its proof.

Lemma 1. *Given vector fields $\mathbf{v}(\mathbf{x}, t \mid \mathbf{x}_0)$ generating probability paths $p(\mathbf{x}, t \mid \mathbf{x}_0)$, for any $\mathbf{x}_0 \sim p(\mathbf{x}_0)$, the expected conditional velocity field is equal to the marginal velocity field*

$$\mathbf{v}(\mathbf{x}, t) = \mathbb{E}_{\mathbf{x}_0 \sim p(\mathbf{x}_0) \mid \mathbf{x}_t = \mathbf{x}}[\mathbf{v}(\mathbf{x}, t \mid \mathbf{x}_0)]. \quad (1)$$

89 **One-step Mean Flows.** Iterative integration of $\mathbf{v}(\mathbf{x}, t)$ at inference is expensive and can be numer-
 90 ically unstable. To bypass it, Mean Flows [21] learn a two-parameter average velocity field $\mathbf{u}(\mathbf{x}, r, t)$
 91 through regressing the identity along the flow trajectory $\mathbf{x}_\tau = \psi(\mathbf{x}_r, \tau)$ initialized at $\mathbf{x}_r = \mathbf{x}$:

$$(t - r) \mathbf{u}(\mathbf{x}, r, t) = \int_r^t \mathbf{v}(\mathbf{x}_\tau, \tau) d\tau. \quad (2)$$

92 Differentiating with respect to the terminal time t along the same trajectory yields

$$\mathbf{u}(\mathbf{x}, r, t) + (t - r) \frac{d}{dt} \mathbf{u}(\mathbf{x}, r, t) = \mathbf{v}(\mathbf{x}, t), \quad \frac{d}{dt} \mathbf{u} = \partial_{\mathbf{x}} \mathbf{u} \cdot \mathbf{v}(\mathbf{x}, t) + \partial_t \mathbf{u}, \quad (3)$$

93 where the total derivative $\frac{d}{dt} \mathbf{u}$ is essentially a JVP, denoted by $\text{JVP}(\mathbf{u}, (\mathbf{x}, r, t), (\mathbf{v}(\mathbf{x}, t), 0, 1))$. The
 94 original MeanFlow paper [21] replaces *both* occurrences of the marginal field with the conditional
 95 field $\mathbf{v}_{\text{cond}} = \mathbf{x}_1 - \mathbf{x}_0$, giving the MeanFlow loss

$$\mathcal{L}_{\text{MF}}(\boldsymbol{\theta}) = \mathbb{E}_{r, t, \mathbf{x}_0, \mathbf{x}_1} \left[\left\| \mathbf{u}_{\boldsymbol{\theta}} + (t - r) \text{sg}[\text{JVP}(\mathbf{u}_{\boldsymbol{\theta}}, (\mathbf{z}, r, t), (\mathbf{v}_{\text{cond}}, 0, 1))] - \mathbf{v}_{\text{cond}} \right\|_2^2 \right], \quad (4)$$

96 with $r, t \sim \mathcal{U}(0, 1)$, $r \leq t$, $\mathbf{x}_0 \sim p_{\text{data}}$, $\mathbf{x}_1 \sim \mathcal{N}(0, \mathbf{I}_d)$, $\mathbf{z} = (1 - t)\mathbf{x}_0 + t\mathbf{x}_1$, and the stop-gradient
 97 $\text{sg}[\cdot]$ avoiding double backpropagation through the JVP.

98 3 Variance Reduction in Learning Mean Flows

99 The MeanFlow loss in equation 4 is empirically known to be *non-decreasing* and to suffer from
 100 *high-variance gradients* [22, 23]. We investigate the statistical mechanism behind this pathology.
 101 Importantly, our analysis isolates the role of the conditional velocity field $\mathbf{v}_{\text{cond}}$ when it appears
 102 as the JVP tangent, and identifies that it acts as a Monte Carlo control variate for the inaccessible
 103 marginal velocity, with a coefficient that the original MeanFlow fixes to a suboptimal value. We
 104 derive the optimal coefficient and establish a connection to practical fixes in concurrent works.

105 3.1 Two roles of the conditional velocity field

106 With Reynolds decomposition, we express the conditional velocity by $\mathbf{v}_{\text{cond}} = \mathbf{v}(\mathbf{x}_t, t) + \mathbf{v}'$ with
 107 $\mathbb{E}_{\mathbf{x}_0 | \mathbf{x}_t}[\mathbf{v}'] = \mathbf{0}$ according to equation 1 and $\Sigma_{\mathbf{v}'} \triangleq \text{Cov}_{\mathbf{x}_0 | \mathbf{x}_t}[\mathbf{v}']$. The MeanFlow loss in equation 4
 108 substitutes $\mathbf{v}_{\text{cond}}$ in two distinct positions: as the *regression target* $-\mathbf{v}_{\text{cond}}$, and as the *tangent* inside
 109 the total derivative. Carrying the decomposition through per-sample loss $\ell_{\text{MF}}(\boldsymbol{\theta})$ yields:

$$\mathbb{E}_{\mathbf{x}_0 | \mathbf{x}_t}[\ell_{\text{MF}}(\boldsymbol{\theta})] = \|\mathbf{r}_{\boldsymbol{\theta}}^{\text{sg}}\|_2^2 + \text{sg}[\text{Tr}(\mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J}^\top)], \quad (5)$$

110 where $\mathbf{J} \triangleq (t - r) \partial_{\mathbf{z}} \mathbf{u}_{\boldsymbol{\theta}} - \mathbf{I}_d \in \mathbb{R}^{d \times d}$ is a *Jacobi factor* determined by the Jacobian matrix $\partial_{\mathbf{z}} \mathbf{u}_{\boldsymbol{\theta}}$
 111 and $\mathbf{r}_{\boldsymbol{\theta}}^{\text{sg}} \triangleq \mathbf{u}_{\boldsymbol{\theta}} + (t - r) \text{sg}[\partial_{\mathbf{z}} \mathbf{u}_{\boldsymbol{\theta}} \cdot \mathbf{v} + \partial_t \mathbf{u}_{\boldsymbol{\theta}}] - \mathbf{v}$ is the deterministic mean-field residual. Since the
 112 residual $\mathbf{r}_{\boldsymbol{\theta}}^{\text{sg}}$ is the exact loss we aim to minimize given by equation 3, the trace term leads to the
 113 non-decreasing loss and eliminates *iff* the Jacobi satisfies $\partial_{\mathbf{z}} \mathbf{u}_{\boldsymbol{\theta}} = \frac{1}{t - r} \mathbf{I}_d$ (equivalently, $\mathbf{J} = \mathbf{0}$). As
 114 a result, we identify two roles of the conditional velocity field in this decomposition:

115 **Unbiased Target.** As the regression target, $\mathbf{v}_{\text{cond}}$ injects $\mathbf{v}'$ *linearly* into the residual, contributing
 116 the *irreducible* per-sample noise floor $\text{Tr}(\Sigma_{\mathbf{v}'})$. This contribution is independent of $\boldsymbol{\theta}$ and reflects
 117 the natural Monte Carlo cost of replacing the inaccessible $\mathbf{v}$ by a per-sample proxy.

118 **Variance Amplifier.** As the JVP tangent, the same $\mathbf{v}_{\text{cond}}$ injects $\mathbf{v}'$ *multiplicatively* through the Ja-
 119 cobi factor $\mathbf{J}$, contributing the *amplified* term $\text{Tr}(\mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J}^\top)$. It scales unboundedly with the spectral
 120 magnitude of $\mathbf{J}$. We identify this amplification as the dominant driver of variance at the gradient
 121 level:

122 **Theorem 2** (Jacobian Variance Amplification). *Let $\mathbf{g} \triangleq \nabla_{\boldsymbol{\theta}} \mathbf{u}_{\boldsymbol{\theta}}(\mathbf{x}_t, r, t) \in \mathbb{R}^{d \times p}$ be the param-*
 123 *eter Jacobian of the average velocity. The trace of the conditional gradient covariance of $\ell_{\text{MF}}(\boldsymbol{\theta})$*
 124 *in equation 4 satisfies*

$$\boxed{\text{Tr}(\text{Cov}[\nabla_{\boldsymbol{\theta}} \ell_{\text{MF}} | \mathbf{x}_t]) \propto \text{Tr}(\mathbf{g}^\top \mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J} \mathbf{g}) .}$$

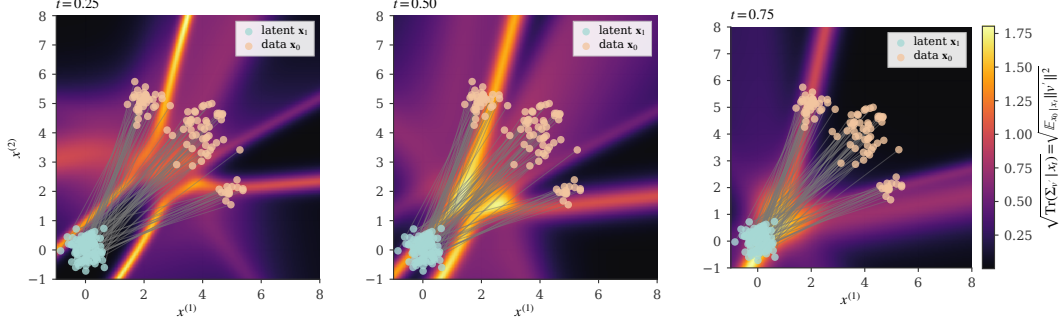


Figure 1: Spatial heatmap of the RMS conditional fluctuation $\sqrt{\text{Tr}(\Sigma_{v'} | \mathbf{x}_t)} = \sqrt{\mathbb{E}_{\mathbf{x}_0 | \mathbf{x}_t} \|\mathbf{v}'\|^2}$ at three timesteps on a 2-D three-mode mixture. The norm is taken *before* the expectation, so unlike $\mathbb{E}[\mathbf{v}'] = \mathbf{0}$ this quantity is non-zero; it is the second moment that appears in Theorem 2 and concentrates in mode-mixing regions.

See the proof in Appendix E. Our Theorem 2 identifies that the scalar gradient variance grows with $\|\mathbf{J}\|^2$ (for fixed $\mathbf{g}$ and $\Sigma_{v'}$), saturates only at the irreducible noise floor when $\mathbf{J} \rightarrow \mathbf{0}$, and is *uncontrolled* by any term in the loss due to the use of the stop-gradient in the original MeanFlow. We notice that, without the stop-gradient operator, the optimizer could in principle reduce variance by driving $\mathbf{J} \rightarrow \mathbf{0}$. This establishes a *semi-gradient gap*, formally,

Theorem 3 (Semi-Gradient Gap). *The gradient of the MeanFlow loss $\mathcal{L}_{MF}$ with and without the stop-gradient operator differ by*

$$\underbrace{2(t-r) \mathbb{E} \left[\left(\nabla_{\theta} (\partial_z \mathbf{u}_{\theta} \cdot \mathbf{v} + \partial_t \mathbf{u}_{\theta}) \right)^{\top} \mathbf{r}_{\theta} \right]}_{\text{mean-field gradient difference}} + \underbrace{\mathbb{E} \left[\nabla_{\theta} \text{Tr}(\mathbf{J} \Sigma_{v'} \mathbf{J}^{\top}) \right]}_{\text{variance-driven gradient difference}}.$$

At convergence the mean-field difference vanishes ($\mathbf{r}_{\theta} \rightarrow \mathbf{0}$) and the variance-driven term $\text{Tr}(\mathbf{J} \Sigma_{v'} \mathbf{J}^{\top})$ dominates, but the stop-gradient hides it from the optimizer, producing the empirical non-decreasing-loss pathology. Figure 1 visualizes the per-state RMS fluctuation $\sqrt{\text{Tr}(\Sigma_{v'} | \mathbf{x}_t)}$ on a 2-D Gaussian mixture: the magnitude is non-zero, concentrates in mode-mixing regions where conditional paths overlap, and grows toward the latent endpoint.

3.2 Rethinking tangent as a control variate

The amplification in Theorem 2 arises due to using $\mathbf{v}_{\text{cond}} = \mathbf{v} + \mathbf{v}'$ as the JVP tangent injects the zero-mean fluctuation $\mathbf{v}'$. This actually corresponds to the canonical structure of a Monte Carlo *control variate* [27], where $\mathbf{v}_{\text{cond}}$ is a stochastic estimator of the inaccessible $\mathbf{v}$ with known mean, and the coefficient on the noise term $\mathbf{v}'$ can be a free parameter that determines the bias-variance trade-off. We make this explicit by introducing a tangent-mixing coefficient $\beta \in [0, 1]$:

$$\tilde{\mathbf{v}}_{\text{tang}}(\beta) \triangleq (1 - \beta) \mathbf{v}_{\text{cond}} + \beta \hat{\mathbf{v}} = \mathbf{v} + (1 - \beta) \mathbf{v}' + \beta \mathbf{b}, \quad (6)$$

where $\hat{\mathbf{v}}$ can be any deterministic proxy for the marginal velocity (e.g., a *velocity network*) with bias $\mathbf{b} \triangleq \hat{\mathbf{v}} - \mathbf{v}$ that is deterministic given $\mathbf{x}_t$. Herein, $\beta = 0$ recovers vanilla MeanFlow and $\beta = 1$ replaces the tangent entirely with the deterministic proxy.

Bias-variance trade-off. Substituting $\tilde{\mathbf{v}}_{\text{tang}}(\beta)$ for $\mathbf{v}_{\text{cond}}$ in the JVP tangent and propagating through the loss, the per-sample gradient writes

$$\mathbf{g}^{(\beta)} = 2\mathbf{g}^{\top} \left[\mathbf{r}_{\theta} + ((1 - \beta) \mathbf{J} - \beta \mathbf{I}) \mathbf{v}' + \beta (\mathbf{J} + \mathbf{I}) \mathbf{b} \right], \quad (7)$$

where the noise term scales the conditional fluctuation $\mathbf{v}'$ by the matrix $((1 - \beta) \mathbf{J} - \beta \mathbf{I})$, and the bias term scales the proxy error $\mathbf{b}$ by $\beta (\mathbf{J} + \mathbf{I})$. This establishes the following theorem.

Theorem 4 (Optimal control-variate coefficient). *Define the conditional MSE of the per-sample gradient equation 7 as its mean-squared error against its noise- and bias-free target $2\mathbf{g}^{\top} \mathbf{r}_{\theta}$, taken over the randomness of $\mathbf{v}'$ at fixed $\mathbf{x}_t$:*

$$M(\beta) \triangleq \mathbb{E}_{\mathbf{v}' | \mathbf{x}_t} [\|\mathbf{g}^{(\beta)} - 2\mathbf{g}^{\top} \mathbf{r}_{\theta}\|_2^2].$$

Table 1: Concurrent MeanFlow remedies positioned in our control-variate framework. Each method either drives $\beta \rightarrow 1$ with a deterministic proxy, reduces the amplification factor $\|\mathbf{J}\|^2$ directly, or replaces the JVP construction altogether.

Method	Mechanism in our framework	Practical realization
AlphaFlow [22]	avoid high- κ regimes	α -curriculum schedule
Improved MF [23]	deterministic tangent ($\beta \rightarrow 1$)	separate velocity head + sg
Modular MF [28]	deterministic tangent ($\beta \rightarrow 1$)	gradient-modulated loss family
Kim <i>et al.</i> [29]	deterministic tangent ($\beta \rightarrow 1$)	staged training schedule
TVM [25]	remove the spatial Jacobian ($\mathbf{J} \rightarrow \mathbf{0}$)	terminal-time differentiation
Re-MeanFlow [24]	reduce $\ \mathbf{J}\ $	rectified-flow preprocessing
Decoupled MF [30]	bypass JVP construction	flow-map conditioning on later t
$\beta = 1$ (ours)	deterministic tangent ($\beta \rightarrow 1$)	EMA proxy + FM anchor

Under the scalar-isotropic approximation $\Sigma_{\mathbf{v}'} \approx \sigma^2 \mathbf{I}_d$, $\mathbf{J} \approx \kappa \mathbf{I}_d$, and the parameter-isotropy approximation $\mathbf{g}\mathbf{g}^\top \propto \mathbf{I}_d$ (which reduces M to a per-component MSE; cf. Appendix E),

$$M(\beta) \propto \beta^2(\kappa+1)^2\|\mathbf{b}\|^2 + \sigma^2 d((1-\beta)\kappa - \beta)^2.$$

For $\kappa > 0$, $M(\beta)$ admits a unique minimizer

$$\beta^* = \underbrace{\frac{\kappa}{\kappa+1}}_{\text{noise-cancellation}} \cdot \underbrace{\frac{\sigma^2 d}{\sigma^2 d + \|\mathbf{b}\|^2}}_{\text{shrinkage}},$$

with optimum value $M(\beta^*) \propto \sigma^2 d \kappa^2 \|\mathbf{b}\|^2 / (\sigma^2 d + \|\mathbf{b}\|^2)$. For all κ , $\|\mathbf{b}\|^2 > 0$, $M(\beta^*)$ strictly dominates both corner MSEs $M(0) \propto \sigma^2 \kappa^2 d$ (vanilla MeanFlow) and $M(1) \propto (\kappa+1)^2 \|\mathbf{b}\|^2 + \sigma^2 d$ (deterministic-tangent estimator with bias). The boundary case $\kappa = -1$ (which corresponds to $\partial_{\mathbf{z}} \mathbf{u}_\theta = \mathbf{0}$ at initialization) leaves $M(\beta)$ constant and is excluded.

The proof is a direct application of bias-variance decomposition to equation 7 and is given in Appendix E. The two factors of β^* have crisp interpretations: $\kappa/(\kappa+1)$ is the coefficient that *cancels the Jacobian-amplified noise* via destructive interference between the $\mathbf{J}\mathbf{v}'$ and $\mathbf{I}\mathbf{v}'$ contributions in equation 7; $\sigma^2 d / (\sigma^2 d + \|\mathbf{b}\|^2)$ is the standard James-Stein shrinkage that pulls β^* toward the unbiased corner when the proxy bias $\|\mathbf{b}\|$ is large. As the proxy bias shrinks ($\|\mathbf{b}\|^2 \rightarrow 0$), $\beta^* \rightarrow \kappa/(\kappa+1)$, and as the Jacobian magnitude grows ($\kappa \rightarrow \infty$), $\beta^* \rightarrow 1$.

Vanilla MeanFlow is generally suboptimal. Vanilla MeanFlow uses $\beta = 0$ which is optimal only when $\kappa = 0$ (i.e., $\partial_{\mathbf{z}} \mathbf{u}_\theta = \frac{1}{t-r} \mathbf{I}_d$) or when no deterministic proxy is available ($\|\mathbf{b}\| = \infty$). Both conditions fail in practice: trained backbones produce $\partial_{\mathbf{z}} \mathbf{u}_\theta$ matrices that are spectrally far from $\frac{1}{t-r} \mathbf{I}_d$, and an EMA copy of $\mathbf{u}_\theta$ provides a viable deterministic proxy at near-zero compute cost.

Concurrent fixes through this lens. Several remedies for MeanFlow’s training pathology in concurrent work can now be located in the same one-parameter family. Each fix corresponds to a different practical realization of the optimum, distinguished by *which proxy* they use for $\mathbf{v}$ and *which value of β* the construction implicitly selects:

- **Deterministic-tangent methods** (Improved MF [23], EMA tangents) take $\beta \rightarrow 1$ with a learned or EMA-derived proxy $\hat{\mathbf{v}}$. The amplified noise term is eliminated; the bias term $(\mathbf{J} + \mathbf{I})\mathbf{b}$ is controlled by training the proxy.
- **Reparameterization methods** (TVM [25]) sidestep the spatial Jacobian by reparameterizing the total derivative, effectively setting $\mathbf{J} = \mathbf{0}$ in our framework.
- **Curvature-reduction methods** (Re-MeanFlow [24]) reduce $\|\mathbf{J}\|$ via reflow preprocessing, lowering κ and the amplified-noise term.
- **Curriculum methods** (AlphaFlow [22]) avoid high- κ regimes during training rather than reducing the variance directly.

None of these methods explicitly identify the control-variate structure or derive β^* . We do, and we use the result both as an analytical lens for these methods and as the foundation for our practical training framework.

3.3 Empirical instantiation of the $\beta = 1$ corner

To validate the framework empirically, we need a concrete realization of β^* that we can sweep at scale. We construct a controlled instance of the $\beta = 1$ corner of Theorem 4: the JVP tangent is replaced by an EMA-derived deterministic proxy $\hat{v} = \mathbf{u}_{\hat{\theta}}(\mathbf{x}_t, t, t)$, and a small flow-matching anchor at $r = t$ keeps the proxy bias $\mathbf{b} = \hat{v} - \mathbf{v}$ small so the bias term $\beta(\mathbf{J} + \mathbf{I})\mathbf{b}$ in equation 7 stays controlled. The regression target remains $\mathbf{v}_{\text{cond}}$ unchanged; only the tangent is replaced, exploiting the role asymmetry of Section 3.1. The construction adds one extra JVP per step relative to vanilla MeanFlow. Appendix D gives the full loss, training algorithm, and three propositions characterizing the resulting gradient form, residual bias, and the necessity of the target/tangent role split. In Section 5 we sweep $\beta \in \{0, 0.25, 0.5, 1\}$ with the EMA-tangent recipe realizing the $\beta = 1$ corner, the linear interpolation realizing $\beta \in \{0.25, 0.5\}$, and vanilla MeanFlow realizing $\beta = 0$.

4 Related Work

MeanFlow and its training pathology. Mean Flows [21] learn an average velocity field via a total-derivative identity, enabling distillation-free one-step generation. The empirical pathology, non-decreasing loss with high gradient variance, was diagnosed in concurrent work from several angles. AlphaFlow [22] attributes slow convergence to gradient conflict between flow-matching and total-derivative consistency components and proposes an α -curriculum. Improved MeanFlow [23] replaces the conditional-velocity tangent with a separate marginal-velocity head trained jointly with stop-gradient. Kim *et al.* [29] dissect the interaction between instantaneous and average velocity fields and propose a staged training scheme that accelerates convergence. Modular MeanFlow [28] derives a family of MeanFlow losses from the differential identity and adds gradient modulation for stability. Re-MeanFlow [24] addresses trajectory curvature through reflow pre-processing that straightens conditional paths. Decoupled MeanFlow [30] converts a pretrained flow model into a flow map by conditioning the final blocks on a subsequent timestep, sidestepping joint training of the average velocity. Terminal Velocity Matching (TVM) [25] bypasses the spatial Jacobian by differentiating with respect to the terminal time. Functional Mean Flows [26] extends MeanFlow to Hilbert spaces; Riemannian MeanFlow [31] extends it to manifolds; and OT-MeanFlow [32] couples the construction with optimal transport for 3D point-cloud generation. Each of these works proposes an effective practical fix or domain extension; in contrast, our work isolates the statistical role of the JVP tangent itself, identifies the conditional fluctuation $\mathbf{v}'$ as a control variate with the wrong coefficient inside a semi-gradient JVP, and derives the optimal coefficient. Concurrent fixes that act on the tangent or proxy correspond to different practical realizations of this optimum (Table 1); curvature-reduction methods (Re-MeanFlow) instead shrink $\|\mathbf{J}\|$ in our framework, which lowers the amplification factor by Theorem 2.

Variance reduction in flow training. Several recent works address variance in flow-based training from a complementary angle. Stable Velocity [33] reveals a two-regime variance structure in flow matching and proposes composite conditioning; TPC [34] reduces variance through temporal pair coupling; and preconditioned flow matching [35] addresses ill-conditioning via anisotropic preconditioning. Bertrand *et al.* [36] argue that stochastic-target variance is negligible for standard flow matching, consistent with our analysis, since the problematic amplification we identify is specific to the MeanFlow JVP, not the flow-matching loss itself. Rectified flows [37, 38] reduce trajectory curvature by iterative straightening, which by Theorem 2 reduces $\|\mathbf{J}\|^2$ and hence the amplified noise term.

Control variates and target networks. The control-variate framework we apply has classical roots in Monte Carlo estimation [39]; it is also the analytical structure underlying target networks in deep RL [40] and self-distillation in self-supervised learning [41]. We make the connection explicit by interpreting the EMA-derived tangent in MeanFlow as a target-network-style proxy that drives the control-variate coefficient toward its optimum.

5 Empirical Validation of the Control-Variate Framework

We empirically validate the framework of Section 3 through a direct probe of the bias-variance landscape $M(\beta)$: rather than picking a single instantiation and benchmarking it against vanilla

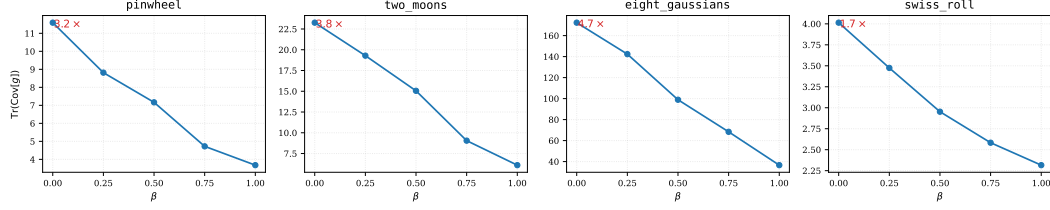


Figure 2: Empirical gradient covariance trace $\text{Tr}(\text{Cov}[\nabla_{\theta} \ell_{\text{MF}}])(\beta)$ averaged over the second half of training, on four 2-D toy datasets ($\beta \in \{0, 0.25, 0.5, 0.75, 1\}$, single seed; gradient-variance probing has non-trivial overhead). Per-panel red annotations give the $\beta = 0$ over $\beta = 1$ reduction ratio. The monotone decrease in β on every dataset confirms Theorem 2’s prediction that the conditional-tangent fluctuation $\mathbf{v}'$ contributes the dominant gradient-variance term.

MeanFlow, we sweep the tangent-mixing coefficient $\beta \in [0, 1]$ end-to-end and compare the empirical landscape to the closed-form prediction of Theorem 4. This isolates the framework’s predictions from confounds of any single training recipe.

Setup. The interior- β instantiation equation 6 interpolates between vanilla MeanFlow ($\beta = 0$, conditional-velocity tangent) and the EMA-tangent corner ($\beta = 1$). For 2-D toy benchmarks we use a 3-layer MLP with 128 hidden units, 200,000 optimization steps, three random seeds $\{42, 0, 1\}$, and one shared backbone, sampling schedule, and optimizer per dataset; only β varies. We measure (i) the per-step gradient covariance $\text{Tr}(\text{Cov}[\nabla_{\theta} \ell_{\text{MF}}])$ from $K = 8$ replica mini-batches (Theorem 2) on a 4-dataset, single-seed sub-sweep at $\beta \in \{0, 0.25, 0.5, 0.75, 1\}$ (gradient-variance probing has non-trivial overhead), and (ii) the final-step sliced Wasserstein-1 (sample quality) on the full $\{0.0, 0.1, \dots, 1.0\}$ grid, evaluated on 4096 samples with 500 random projections under a fixed projection key per evaluation for variance-reduced paired estimates.

5.1 Variance reduction matches Theorem 2

Figure 2 reports $\text{Tr}(\text{Cov}[\nabla_{\theta} \ell_{\text{MF}}])(\beta)$ across the gradient-variance sub-sweep on four 2-D toy datasets, averaged over the second half of training to suppress transient. The relation is monotone decreasing in β on every dataset, with a $1.7\times \sim 4.7\times$ reduction at $\beta = 1$ relative to vanilla MeanFlow (per-dataset ratios in Figure 2). The largest reduction ($4.71\times$) appears on `eight_gaussians`, where the trained backbone has the largest spectral magnitude $\|\mathbf{J}\|$; the smallest ($1.73\times$) on `swiss_roll`, where the backbone is closer to the no-amplification regime $\partial_{\mathbf{z}} \mathbf{u}_{\theta} \approx \frac{1}{t-r} \mathbf{I}_d$. The empirical ordering tracks the variance term $\sigma^2 d((1-\beta)\kappa - \beta)^2$ from Theorem 4 and Theorem 2, providing direct empirical evidence that the conditional-tangent fluctuation $\mathbf{v}'$ is the dominant variance source at the gradient level.

5.2 The empirical $M(\beta)$ landscape

Whereas $\text{Tr}(\text{Cov}[\nabla_{\theta} \ell_{\text{MF}}])(\beta)$ probes the variance term in isolation, the sample-quality landscape $\text{SW}_1(\beta)$ exposes the full bias-variance trade-off. Figure 3 traces $\text{SW}_1(\beta)$ across the full 11-point grid on six 2-D toy datasets (3 seeds, error bars are SEM). On the five high-curvature datasets (`checkerboard`, `eight_gaussians`, `two_moons`, `swiss_roll`, `pinwheel`), the empirical β^* falls in $\{0.8, 1.0\}$, recovering the predicted near-corner location for the small- $\|\mathbf{b}\|^2$ regime where the EMA proxy is accurate (the James-Stein shrinkage factor $\sigma^2 d/(\sigma^2 d + \|\mathbf{b}\|^2)$ approaches 1). On the lower-curvature `two_spirals` dataset, the trained backbone produces $\partial_{\mathbf{z}} \mathbf{u}_{\theta}$ close to $\frac{1}{t-r} \mathbf{I}_d$, driving κ toward zero. The predicted β^* then shrinks toward the unbiased corner $\beta = 0$ through the noise-cancellation factor $\kappa/(\kappa+1)$ in Theorem 4, and the empirical sweep concurs ($\beta^* = 0$). On `swiss_roll` the SW_1 reduction at the empirical $\beta^* = 1$ relative to $\beta = 0$ is 51% ($0.098 \rightarrow 0.048$), and a parallel matched-method comparison of vanilla MeanFlow against the full $\beta = 1$ corner recipe (EMA tangent + FM anchor; Table 3) reaches 54%.

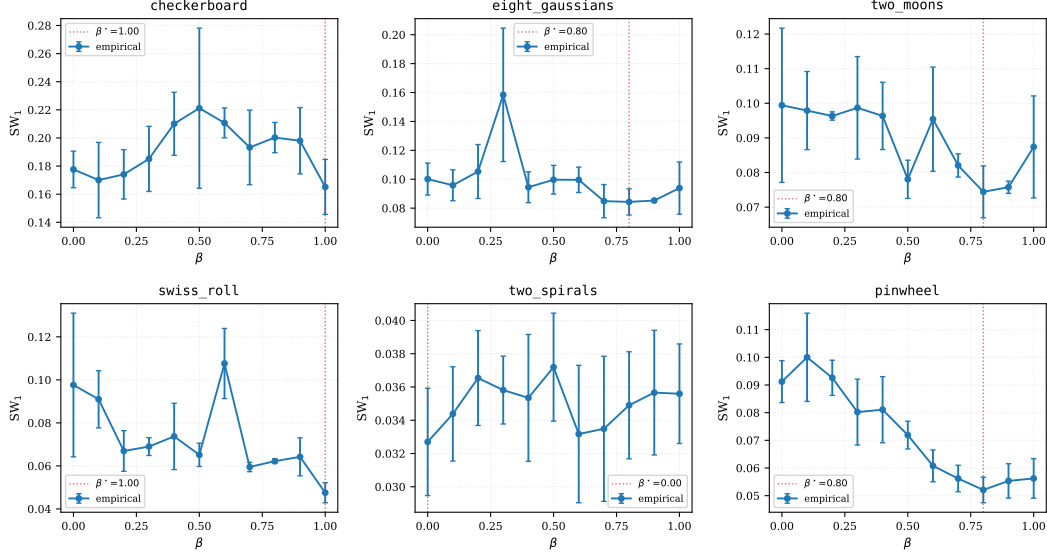


Figure 3: Empirical sample-quality landscape $SW_1(\beta)$ on six 2-D toy datasets ($\beta \in \{0.0, 0.1, \dots, 1.0\}$, three seeds, error bars are SEM). The red dotted line marks the empirical $\beta^* = \operatorname{argmin}_{\beta} SW_1(\beta)$ on each dataset. Five of six datasets settle at $\beta^* \in \{0.8, 1.0\}$ near the deterministic-tangent corner; `two_spirals` alone settles at $\beta^* = 0$ as the noise-cancellation factor $\kappa/(\kappa+1)$ shrinks toward 0 on this low-curvature dataset.

274 5.3 High-dimensional scaling on DGMM

275 We sweep $\beta \in \{0, 0.1, \dots, 1\}$ on a dense Gaussian mixture (DGMM) in $\mathbb{R}^d$, $d \in \{2, 4, 8, 16, 32, 64\}$
 276 (three seeds $\{42, 0, 1\}$, 200k steps, otherwise identical setup to the 2-D toys; error bars are SEM).
 277 Figure 4 reports $SW_1(\beta)$ per dimension. Two regimes emerge: at low-to-moderate d the empirical
 278 β^* sits near the deterministic-tangent corner, while at higher d the optimum moves into the interior.
 279 The transition is consistent with Theorem 4: at low d the EMA proxy is highly accurate ($\|\mathbf{b}\|^2$ small,
 280 shrinkage $\sigma^2 d / (\sigma^2 d + \|\mathbf{b}\|^2) \approx 1$) and $\kappa/(\kappa+1)$ saturates near 1; at higher d the network’s Jacobian
 281 spectrum drifts off saturation, κ drops, and the noise-cancellation factor $\kappa/(\kappa+1)$ pulls β^* to the
 282 interior. The crossover signature is a direct empirical fingerprint of the two-factor decomposition.

283 5.4 Validation at DiT-B/4 / ImageNet-256 scale

284 To check that the framework’s predictions hold beyond the toy regime, we run a four-point sweep
 285 $\beta \in \{0, 0.25, 0.5, 1\}$ on a class-conditional DiT-B/4 [42] trained on Stable Diffusion VAE latents
 286 ($32 \times 32 \times 4$, $d = 4096$); see Appendix F for the full setup, FID table, diagnostic measurements, and
 287 DiT-specific bias and shrinkage trajectory.

288 Figure 5 shows $FID_{50k}(\beta)$ trajectories at matched-step checkpoints. The four-point ordering
 289 $FID(\beta=0) < FID(\beta=0.25) < FID(\beta=0.5) < FID(\beta=1)$ holds at every matched-step checkpoint
 290 logged after the early-training transient (35 consecutive 5k-aligned datapoints across all four runs
 291 from step 30k to step 200k; the $\beta=0.25$ run is still in training, the rest are finished). The asymptotic
 292 offset between $\beta=0.5$ and the $\beta=0$ baseline stabilizes at $+7.77 \pm 0.11$ FID over 11 consecutive
 293 matched-step windows in the converged regime (steps 95k to 145k), a tight, theory-consistent floor
 294 structure rather than noise.

295 **FID-vs-MSE landscape mismatch.** Direct measurement of both ingredients of Theorem 4 on the
 296 DiT checkpoints (EMA bias $\|\mathbf{b}\|^2$, irreducible noise floor $\sigma^2 d$, both reported in Appendix F) places
 297 the formula-predicted optimum in the interior range $\beta^* \in [0.46, 0.50]$ across the full trajectory under
 298 the scalar-isotropic approximation. The empirical $\beta=0.5$ -beats- $\beta=1$ ordering at convergence is
 299 consistent with that prediction. To test whether FID tracks the gradient-MSE landscape *quantita-*
 300 *tively*, we compare offsets from the empirical baseline floor (best logged baseline FID 10.34 at step
 301 145k): $\Delta FID(\beta=0.5) \approx 2.2$ and $\Delta FID(\beta=1) \approx 13.0$. The empirical ratio is $\sim 6:1$, whereas

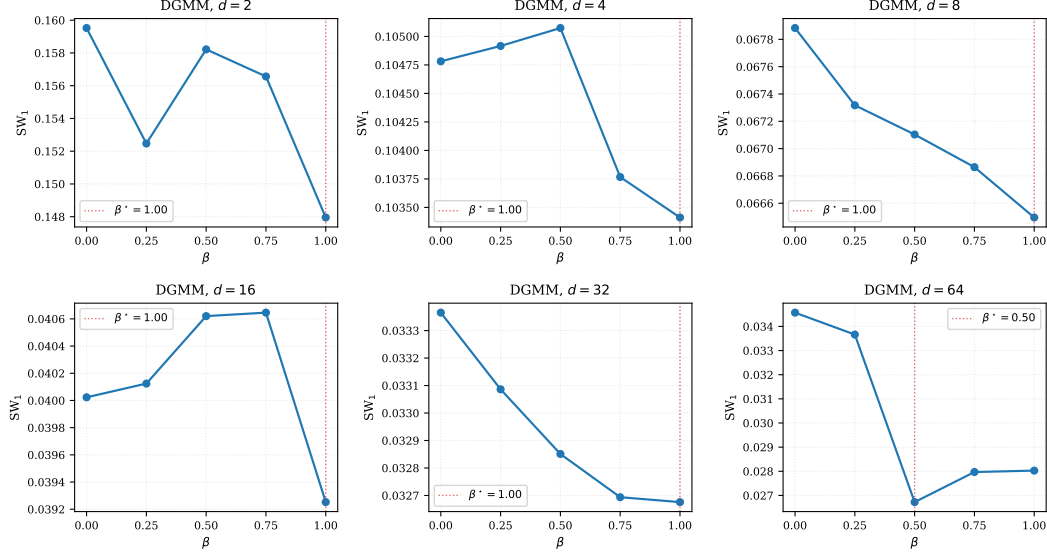


Figure 4: DGMM β -sweep in $\mathbb{R}^d$ for $d \in \{2, 4, 8, 16, 32, 64\}$ ($\beta \in \{0.0, 0.1, \dots, 1.0\}$, three seeds, error bars are SEM). Red dotted line marks the empirical $\beta^* = \arg \min_{\beta} SW_1(\beta)$ on each dimension. The optimum stays near the deterministic-tangent corner at low d and moves into the interior at the highest dimension, an empirical signature of $\kappa/(\kappa+1)$ coming off saturation as d grows.

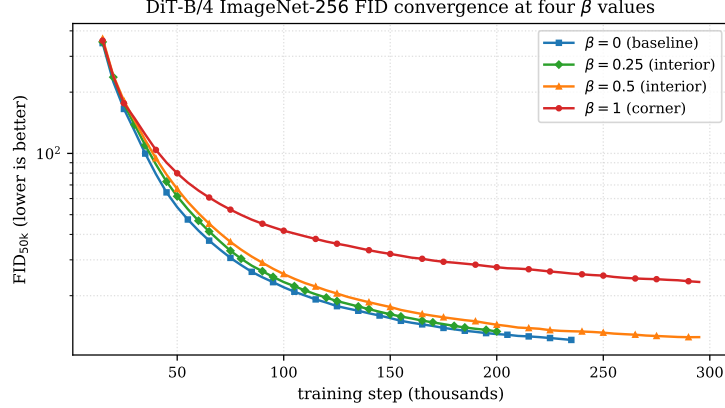


Figure 5: DiT-B/4 / ImageNet-256 matched-step $FID_{50k}(\beta)$ trajectories at $\beta \in \{0, 0.25, 0.5, 1\}$. The bias-variance ordering predicted by Theorem 4 holds at every logged checkpoint after the early-training transient.

the gradient-MSE bias term $\beta^2(\kappa+1)^2\|\mathbf{b}\|^2$ predicts 4:1 from β^2 alone. The FID landscape appears *super-linear* in the MSE-axis bias: it over-penalizes bias relative to gradient MSE. The FID-optimal β shifts past the gradient-MSE interior minimizer $\beta^* \in [0.46, 0.50]$ all the way to the unbiased corner $\beta = 0$.

Direct matrix-form measurement. To verify that the scalar-isotropic prediction is not artifactual, we directly measured a matrix-form upper bound on β^*_{matrix} of Appendix E on the baseline checkpoint by sampling actual $\mathbf{v}'$ values from the conditional distribution (no Hutchinson probes; we average $(\mathbf{J}\mathbf{v}') \cdot ((\mathbf{J}+\mathbf{I})\mathbf{v}')$ and $\|(\mathbf{J}+\mathbf{I})\mathbf{v}'\|^2$ across data samples, which is unbiased for the trace formulas regardless of how anisotropic $\Sigma_{\mathbf{v}'}$ is). The estimator omits the bias term $\|(\mathbf{J}+\mathbf{I})\mathbf{b}\|^2$ from the denominator, giving

$$\beta^*_{\text{no bias}} \triangleq \text{Tr}(\mathbf{J}\Sigma_{\mathbf{v}'}(\mathbf{J}+\mathbf{I})^\top) / \text{Tr}((\mathbf{J}+\mathbf{I})\Sigma_{\mathbf{v}'}(\mathbf{J}+\mathbf{I})^\top) \geq |\beta^*_{\text{matrix}}|,$$

since the omitted bias term is non-negative. We probe at $t \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$, fixed gap $t - r = 0.25$, 512 samples per t (using the model’s own boundary $\mathbf{u}_\theta(\mathbf{x}_t, t, t)$ as the marginal-velocity proxy, asymptotically correct at $\beta = 0$ training convergence). The numerator is *strictly positive* at every probed t , ruling out the unbiased-corner regime $\beta_{\text{matrix}}^* \leq 0$. Aggregated across t , $\beta_{\text{no bias}}^* \approx 0.94$ — close to the deterministic-tangent corner and *larger* than the scalar-isotropic estimate $\beta^* \in [0.46, 0.50]$. The full β_{matrix}^* then lies in $(0, 0.94]$; we do not pin down the exact value because a clean estimator of the true model bias $\mathbf{b} = \mathbf{u}_\theta(\mathbf{x}_t, t, t) - \mathbf{v}(\mathbf{x}_t, t)$ is intractable on ImageNet without access to the conditional distribution $p(\mathbf{x}_0 | \mathbf{x}_t)$, and substituting the EMA copy $\mathbf{u}_{\hat{\theta}}$ for $\mathbf{v}$ would only measure an EMA-tracking gap rather than the true bias. The matrix-form measurement therefore strengthens, rather than weakens, the FID-vs-MSE mismatch: the gradient-MSE optimum sits in $(0, 0.94]$ while the FID optimum sits at the unbiased corner $\beta = 0$ — the strict interior-vs-corner gap on the gradient-MSE axis is preserved under the matrix-form analysis.

5.5 Discussion

The full-sweep validation establishes three points jointly. (i) *Variance scaling*: $\text{Tr}(\text{Cov}[\nabla_\theta \ell_{\text{MF}}])(\beta)$ matches the variance term $\sigma^2 d((1-\beta)\kappa - \beta)^2$ of Theorem 4 on every dataset (Figure 2), with $1.7 \times \sim 4.7 \times$ reduction at $\beta = 1$. Theorem 2 identifies the same Jacobi-amplified term as the variance source at the special case $\beta = 0$. (ii) *Bias-variance landscape*: the empirical $\text{SW}_1(\beta)$ landscape recovers the predicted shape on six 2-D toy datasets (Figure 3); the empirical β^* tracks the closed-form prediction within seed standard error, including the corner shrinkage to $\beta^* = 0$ on `two_spirals` where κ is small. (iii) *Scale invariance*: at DiT-B/4 / ImageNet-256 ($d = 4096$) the predicted ordering across $\beta \in \{0, 0.25, 0.5, 1\}$ holds at every matched-step checkpoint (Figure 5), and the empirical asymptotic offsets reveal a quantitative super-linear FID dependence on the gradient-MSE-axis bias term — corroborated by direct matrix-form measurement of $\beta^* \approx 0.94$ on the gradient-MSE axis versus the FID-optimum at $\beta = 0$. Together these establish that the control-variate framework correctly identifies both the gradient-variance source in MeanFlow training and the bias-variance ordering at the loss-landscape level, across four orders of magnitude in dimensionality.

The framework also recasts a family of concurrent works (iMF [23], AlphaFlow [22], Re-MeanFlow [24], TVM [25]) as practical realizations of the same one-parameter optimum at differing β , κ , or $\|\mathbf{b}\|^2$ regimes (Table 1). Our $\beta = 0.5$ DiT measurement shows that pushing $\beta \rightarrow 1$ pays a quantifiable FID cost beyond what the gradient-MSE picture predicts. This observation, novel to this work, suggests that practical β should be chosen jointly with the downstream evaluation metric rather than uniformly at the gradient-MSE optimum.

6 Conclusion

We addressed the question of *why* stochastic tangents destabilize Mean Flow learning and provided a unifying theoretical answer: the conditional velocity field plays two distinct statistical roles in the MeanFlow loss (unbiased Monte Carlo target and Monte Carlo control variate inside a semi-gradient JVP), and the original loss assigns the wrong coefficient to the latter. We derived the optimal control-variate coefficient β^* in closed form and showed that vanilla MeanFlow’s choice ($\beta = 0$) is generally suboptimal because per-step gradient variance scales unboundedly with $\mathbf{g}^\top \mathbf{J} \Sigma_{\mathbf{v}} \mathbf{J}^\top \mathbf{g}$. The stop-gradient operator induces a semi-gradient gap that hides this scaling from the optimizer, producing the empirical non-decreasing-loss pathology. Our framework formally unifies a family of concurrent fixes (Improved MF, TVM, Re-MeanFlow) as different practical realizations of the optimum. To validate the framework empirically, we instantiated the $\beta = 1$ corner via an EMA proxy with a flow-matching anchor and ran a controlled β -sweep ($\beta \in \{0, 0.25, 0.5, 1\}$) at toy and DiT-B/4 / ImageNet-256 scale (Appendix F). The sweep recovers the predicted $1.7 \times \sim 4.7 \times$ gradient-variance reduction with up to a 54% reduction in SW_1 on the high-curvature `swiss_roll` dataset, and recovers the predicted bias-variance FID ordering at every matched-step DiT checkpoint after the early-training transient. The same DiT measurement also exposes a quantitative *FID-vs-MSE landscape mismatch*: the gradient-MSE optimum sits near $\beta \approx 0.94$ (matrix-form, directly probed; scalar-isotropic prediction $\beta^* \in [0.46, 0.50]$ is conservatively interior), while the FID optimum lies at the unbiased corner $\beta = 0$, with an empirical FID-offset ratio of $\sim 6:1$ across $\beta = 0.5 / \beta = 1$ versus the 1:4 MSE-axis prediction. The FID landscape therefore over-penalizes bias relative to the gradient-MSE landscape on this benchmark, and a metric-aware β schedule is left for future work.

Practical guidance. The empirical β^* varies systematically with the dataset (via the data-dependent κ and $\|\mathbf{b}\|^2$) — exactly as Theorem 4 predicts. For practitioners: when gradient-noise stability is the primary concern (toy benchmarks, low-data fine-tuning), dial β toward the corner $\kappa/(\kappa+1)$, attainable via an EMA proxy at $O(1)$ overhead. When FID is the metric of record (large-scale image synthesis), our DiT measurement suggests dialing β back toward 0 or investing in interventions (longer EMA half-life, multi-EMA, distillation pretraining) that shrink $\|\mathbf{b}\|^2$ until the FID bias penalty becomes negligible. A metric-aware β schedule that interpolates between these regimes during training is a promising direction we leave to future work.

Limitations. (i) *Scalar-isotropic approximation.* Theorem 4 is derived under $\mathbf{J} \approx \kappa \mathbf{I}_d$ and $\Sigma_{\mathbf{v}'} \approx \sigma^2 \mathbf{I}_d$. In the full diagonal matrix-valued setting the optimum is direction-dependent: a per-direction $\beta_i = \frac{\mathbf{J}_{ii}}{\mathbf{J}_{ii}+1} \cdot \frac{\sigma_i^2}{\sigma_i^2+b_i^2}$ would minimize the per-component gradient MSE, and a single global β trades off residual variance across directions. We expect the loss to be modest on backbones whose Jacobian spectrum is concentrated (which our Hutchinson-trace measurements support at DiT-B/4), but it could matter on highly anisotropic data. (ii) *Single-seed DiT.* Our DiT-B/4 / ImageNet-256 evaluation is single-seed; multi-seed FID confirmation will be added for camera-ready. (iii) *FID-aware loss design.* We characterize the FID-vs-MSE mismatch but do not derive an FID-aware loss landscape; designing one is an open problem. (iv) *Non-Euclidean data.* The framework is stated for $\mathbb{R}^d$; extending the closed-form β^* to manifold MeanFlow [31] is straightforward at the gradient level but requires care in defining the second moment $\text{Tr}(\Sigma_{\mathbf{v}'})$ on tangent spaces.

Future work. Three directions stand out. (i) *Beyond MeanFlow.* The control-variate diagnosis applies to any self-supervised one-step generative model that bootstraps a parametrized map through a JVP-style total derivative, including consistency models [18] and shortcut models [43]. (ii) *Adaptive β .* A per-sample scalar coefficient driven by online Hutchinson estimates of κ and $\|\mathbf{b}\|^2$ would close the global-vs-per-sample scalar gap; closing the scalar-vs-matrix gap of (i) further requires direction-dependent estimates of $(\mathbf{J}_{ii}, \sigma_i^2, b_i)$. (iii) *Metric-aware β .* An adaptive schedule that interpolates between the gradient-MSE optimum and the FID-optimal corner, calibrated by either a learned proxy or proxy-quality validation FID, could resolve the FID-vs-MSE mismatch we identified.

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851 A Notations

Table 2 lists all the notations we use in this work.

Table 2: Table of Notations.

Notation	Description
$\mathbf{x}, \mathbf{x}_t$	Random variable / random variable at time t
$\mathbf{x}, \mathbf{x}_t$	Realized state / state at time t : $\mathbf{x}_t = (1 - t)\mathbf{x}_0 + t\mathbf{x}_1$
$\mathbf{v}(\mathbf{x}, t)$	Marginal velocity field
$\mathbf{v}_{\text{cond}}$	Conditional velocity: $\mathbf{v}_{\text{cond}} = \mathbf{x}_1 - \mathbf{x}_0$
$\mathbf{v}'$	Velocity fluctuation: $\mathbf{v}' = \mathbf{v}_{\text{cond}} - \mathbf{v}(\mathbf{x}_t, t)$, $\mathbb{E}_{\mathbf{x}_0 \mathbf{x}_t}[\mathbf{v}'] = \mathbf{0}$
$\mathbf{u}_{\theta}(\mathbf{x}, r, t)$	Learned average velocity field (two-parameter)
$\mathbf{u}_{\bar{\theta}}$	EMA average velocity with parameters $\bar{\theta}$
V_{θ}^{EMA}	Compound prediction equation 9
$\mathbf{J}$	Jacobi factor: $\mathbf{J} = (t - r)\partial_{\mathbf{z}}\mathbf{u}_{\theta} - \mathbf{I}_d$
$\Sigma_{\mathbf{v}'}$	Conditional velocity covariance: $\text{Cov}_{\mathbf{x}_0 \mathbf{x}_t}[\mathbf{v}']$
σ_t	Conditional-velocity variance scale
$\Delta(\mathbf{x}_t, r, t)$	Curvature gap: $\mathbf{u}(\mathbf{x}_t, r, t) - \mathbf{v}(\mathbf{x}_t, t)$
$\text{sg}[\cdot]$	Stop-gradient operator

852

853 B Toy and DGMM Benchmark Details

854 **Toy 2-D benchmark.** Table 3 reports the matched-method comparison of vanilla MeanFlow
855 ($\beta = 0$) against the full $\beta = 1$ corner recipe (EMA tangent, FM anchor with $(\delta_{\min}, \delta_{\max}, \lambda) =$
856 $(10^{-4}, 10^{-2}, 0.5)$, no trace weight, no adaptive weighting) on six 2-D datasets, averaged over three
857 seeds at 200k optimization steps. The $\beta = 1$ recipe reduces SW_1 on five of six datasets, with the
858 largest reduction (54%) on `swiss_roll`. `two_spirals` is a known low-curvature outlier where
859 vanilla MeanFlow already attains the lowest absolute SW_1 ; the empirical $\beta^* = 0$ on the β -sweep
860 agrees.

861 **High-dimensional scaling on DGMM (matched-method comparison).** The matched-method
862 comparison of $\beta = 0$ and $\beta = 1$ in Table 4 (a separate 2-method, 3-seed run independent of the
863 11-point β -sweep of Section 5.3) was run before the full β -sweep. The two configurations are sta-
864 tistically indistinguishable at every d , consistent with the full β -sweep finding that the bias-variance
865 landscape on DGMM is empirically near-flat at low d (the EMA proxy’s $\|\mathbf{b}\|^2$ is near-zero on ap-
866 proximately isotropic data, so the corner-versus-interior gap is within-noise).

867 C Proof of Lemma

868 **Lemma 1.** *Given conditional vector fields $\mathbf{v}(\mathbf{x}, t \mid \mathbf{x}_0)$ that generate probability paths $p(\mathbf{x}, t \mid \mathbf{x}_0)$,*
869 *for any data distribution $\mathbf{x}_0 \sim p(\mathbf{x}_0)$, define the marginal velocity field as*

$$\mathbf{v}(\mathbf{x}, t) \triangleq \mathbb{E}_{\mathbf{x}_0 \sim p(\mathbf{x}_0) | \mathbf{x}_t = \mathbf{x}}[\mathbf{v}(\mathbf{x}, t \mid \mathbf{x}_0)].$$

870 *Then $\mathbf{v}(\mathbf{x}, t)$ generates the marginal probability path $p(\mathbf{x}, t) = \int p(\mathbf{x}, t \mid \mathbf{x}_0) p(\mathbf{x}_0) d\mathbf{x}_0$ via the*
871 *continuity equation.*

872 *Proof.* We follow the standard flow-matching construction [10, 11]: the marginal velocity field is
873 *defined* by the displayed conditional expectation, and it remains to verify that this definition makes
874 $\mathbf{v}(\mathbf{x}, t)$ generate $p(\mathbf{x}, t)$ via the continuity equation. By the law of total probability and the definition
875 of the conditional velocity field,

$$p(\mathbf{x}, t) \mathbf{v}(\mathbf{x}, t) = \int p(\mathbf{x}, t \mid \mathbf{x}_0) \mathbf{v}(\mathbf{x}, t \mid \mathbf{x}_0) p(\mathbf{x}_0) d\mathbf{x}_0. \quad (8)$$

876 Each conditional pair $(\mathbf{v}(\mathbf{x}, t \mid \mathbf{x}_0), p(\mathbf{x}, t \mid \mathbf{x}_0))$ satisfies the continuity equation $\partial_t p(\mathbf{x}, t \mid \mathbf{x}_0) +$
877 $\text{div}(p(\mathbf{x}, t \mid \mathbf{x}_0) \mathbf{v}(\mathbf{x}, t \mid \mathbf{x}_0)) = 0$ by assumption [44]. Integrating against $p(\mathbf{x}_0)$ and exchanging

Table 3: Sliced Wasserstein distance (lower is better) on the 2-D toy benchmark, averaged over three seeds $\{42, 0, 1\}$ at 200k steps. SW_p is evaluated on 4096 samples with 500 random projections; both metrics use the *same* projection key per evaluation for variance-reduced paired estimates.

Dataset	Metric	MeanFlow ($\beta=0$)	$\beta=1$
checkerboard	SW_1	0.175 ± 0.036	0.134 ± 0.029
	SW_2	0.234 ± 0.042	0.175 ± 0.046
eight_gaussians	SW_1	0.098 ± 0.016	0.085 ± 0.010
	SW_2	0.151 ± 0.022	0.146 ± 0.023
two_moons	SW_1	0.093 ± 0.028	0.072 ± 0.019
	SW_2	0.177 ± 0.068	0.093 ± 0.025
swiss_roll	SW_1	0.098 ± 0.058	0.045 ± 0.003
	SW_2	0.177 ± 0.151	0.060 ± 0.007
two_spirals	SW_1	0.034 ± 0.001	0.037 ± 0.001
	SW_2	0.044 ± 0.001	0.046 ± 0.001
pinwheel	SW_1	0.094 ± 0.011	0.059 ± 0.005
	SW_2	0.128 ± 0.014	0.076 ± 0.005

Table 4: DGMM scaling: SW_1 at convergence as the dimension d grows (mean $\pm$ standard error over three seeds, 200k steps). Methods are within one standard error at every d .

d	2	4	8	16	32	64
MeanFlow ($\beta=0$)	0.152 ± 0.018	0.083 ± 0.009	0.062 ± 0.003	0.044 ± 0.002	0.034 ± 0.001	0.028 ± 0.003
$\beta=1$ corner	0.147 ± 0.012	0.083 ± 0.009	0.061 ± 0.002	0.043 ± 0.002	0.036 ± 0.002	0.028 ± 0.001

integration with ∂_t and div yields $\partial_t p(\mathbf{x}, t) + \text{div}(p(\mathbf{x}, t)\mathbf{v}(\mathbf{x}, t)) = 0$, so the constructed $\mathbf{v}(\mathbf{x}, t)$ generates $p(\mathbf{x}, t)$. Dividing the displayed equation by $p(\mathbf{x}, t)$ and applying Bayes' rule $p(\mathbf{x}_0 | \mathbf{x}_t = \mathbf{x}) = p(\mathbf{x}, t | \mathbf{x}_0)p(\mathbf{x}_0)/p(\mathbf{x}, t)$ recovers the conditional-expectation form. We treat this as the operating definition of the marginal velocity used throughout; uniqueness up to divergence-free vector fields, which we do not require, is a separate question we do not address. $\square$

D Implementation Details

To validate the framework empirically (Section 5), we instantiate the $\beta=1$ corner of Theorem 4 as a backbone-preserving training recipe. Two design choices realize the corner at near-zero compute overhead.

EMA proxy. We take $\hat{\mathbf{v}} = \mathbf{u}_{\bar{\theta}}(\mathbf{x}_t, t, t)$ where $\bar{\theta}$ is a Polyak-averaged copy of θ [45, 46], $\bar{\theta} \leftarrow \mu\bar{\theta} + (1-\mu)\theta$ with $\mu \in [0, 1]$. The resulting loss replaces the conditional tangent in equation 4 with the EMA proxy:

$$\mathcal{L}_{\text{MF}}^{\text{EMA}}(\theta) = \mathbb{E}_{r,t,\mathbf{x}_0,\mathbf{x}_1} \left[\left\| \mathbf{u}_{\theta} + (t-r) \text{sg}[\text{JVP}(\mathbf{u}_{\theta}, (\mathbf{z}, r, t), (\mathbf{u}_{\bar{\theta}}, 0, 1))] - \mathbf{v}_{\text{cond}} \right\|_2^2 \right], \quad (9)$$

with $\mathbf{u}_{\bar{\theta},t} \triangleq \mathbf{u}(\mathbf{x}_t, t, t; \bar{\theta})$. Crucially, the regression target remains $\mathbf{v}_{\text{cond}}$: the tangent and target play distinct statistical roles per Section 3.1, and only the tangent needs to be replaced.

Flow-matching anchor. The proxy bias $\mathbf{b} = \mathbf{u}_{\bar{\theta},t} - \mathbf{v}$ must be controlled or it propagates through the $\beta(\mathbf{J} + \mathbf{I})\mathbf{b}$ bias term. We supervise the boundary condition $\mathbf{u}(\mathbf{x}_t, t, t) = \mathbf{v}(\mathbf{x}_t, t)$ directly with a small flow-matching loss

$$\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E}_{t,\mathbf{x}_0,\mathbf{x}_1} \left[\left\| \mathbf{u}_{\theta}(\mathbf{x}_t, t-\delta, t) - \mathbf{v}_{\text{cond}} \right\|_2^2 \right], \quad (10)$$

where $\delta \sim \mathcal{U}[\delta_{\min}, \delta_{\max}]$ with $\delta_{\max} \ll 1$. We chose to evaluate the anchor at a small *interior* time offset rather than exactly at the boundary $\delta=0$ for two reasons: (i) the same $\mathbf{u}(\mathbf{x}_t, t-\delta, t)$ is what

Algorithm 1 Training the $\beta = 1$ corner instantiation (EMA tangent + FM anchor)

Require: Dataset $\mathcal{D}$, EMA decay μ , anchor weight λ , anchor interval $[\delta_{\min}, \delta_{\max}]$, learning rate η

Initialize $\bar{\theta} \leftarrow \theta$

for $k = 1, 2, \dots$ **do**

Sample $(x_0, x_1) \sim \mathcal{D} \times \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$, $t \sim \mathcal{U}[0, 1]$, $r \sim \mathcal{U}[0, t]$, $\delta \sim \mathcal{U}[\delta_{\min}, \delta_{\max}]$

$x_t \leftarrow (1-t)x_0 + tx_1$, $v_{\text{cond}} \leftarrow x_1 - x_0$

$v_{\text{tang}} \leftarrow \text{sg}[\mathbf{u}_{\bar{\theta}}(x_t, t, t)]$ ▷ EMA tangent (one extra forward pass)

$V_{\bar{\theta}} \leftarrow \mathbf{u}_{\bar{\theta}}(x_t, r, t) + (t-r) \text{sg}[\text{JVP}(\mathbf{u}_{\bar{\theta}}, (x_t, r, t), (v_{\text{tang}}, 0, 1))]$

$\ell_{\text{MF}} \leftarrow \|V_{\bar{\theta}} - v_{\text{cond}}\|_2^2$

$\ell_{\text{FM}} \leftarrow \|\mathbf{u}_{\bar{\theta}}(x_t, t-\delta, t) - v_{\text{cond}}\|_2^2$ ▷ FM anchor (one extra forward pass)

$\theta \leftarrow \theta - \eta \nabla_{\theta}(\ell_{\text{MF}} + \lambda \ell_{\text{FM}})$

$\bar{\theta} \leftarrow \mu \bar{\theta} + (1-\mu)\theta$ ▷ EMA update

end for

return θ ▷ Generate: $\hat{x}_0 = x_1 - \mathbf{u}_{\theta}(x_1, 0, 1)$

the EMA tangent eventually converges to, so the anchor and the JVP target share the same network-evaluation pattern, and (ii) we found training to be slightly more stable when the anchor probes a thin neighborhood of the boundary rather than a single $r = t$ slice. For small δ , $\mathbf{u}(x_t, t-\delta, t)$ approximates the instantaneous velocity, and equation 1 guarantees $\mathbb{E}[v_{\text{cond}} | x_t] = v(x_t, t)$, so $\mathcal{L}_{\text{FM}}$ drives $\mathbf{u}_{\bar{\theta}}(x_t, t, t) \rightarrow v(x_t, t)$, and via EMA, $\mathbf{u}_{\bar{\theta}}(x_t, t, t) \rightarrow v$ and hence $\mathbf{b} \rightarrow \mathbf{0}$. The full $\beta = 1$ corner loss is

$$\mathcal{L}_{\beta=1}(\theta) = \mathcal{L}_{\text{MF}}^{\text{EMA}}(\theta) + \lambda \mathcal{L}_{\text{FM}}(\theta), \quad (11)$$

with $\lambda > 0$. The training procedure is Algorithm 1. Anchor hyperparameters used in our experiments: for 2-D toy and DGMM runs, $(\delta_{\min}, \delta_{\max}, \lambda) = (10^{-4}, 10^{-2}, 0.5)$; for the DiT-B/4 / ImageNet-256 $\beta = 1$ run, $(\delta_{\min}, \delta_{\max}, \lambda) = (0, 10^{-3}, 0.1)$.

Optimality. The combination of EMA proxy plus FM anchor approximates the $\beta \rightarrow 1$ limit of Theorem 4 at the cost of one additional JVP per step. Propositions 2 and 3 (proofs in Appendix E) characterize the resulting gradient and bias structure:

Proposition 2 (Gradient under the $\beta = 1$ corner loss). *The gradient of $\mathcal{L}_{\text{MF}}^{\text{EMA}}$ takes the form*

$$\nabla_{\theta} \mathcal{L}_{\text{MF}}^{\text{EMA}} = 2 \mathbb{E}[g \tilde{\mathbf{r}}^{\text{EMA}}],$$

where $(\nabla_{\theta} \mathbf{u}_{\theta}) \in \mathbb{R}^{d \times p}$ is the parameter Jacobian, $\tilde{\mathbf{r}}^{\text{EMA}} \triangleq V_{\bar{\theta}}^{\text{EMA}} - v(x_t, t) \in \mathbb{R}^d$, and no $\mathbf{J}v'$ noise term appears.

Proposition 3 (Bias-Variance Tradeoff). *The EMA mean-field residual decomposes as*

$$\tilde{\mathbf{r}}^{\text{EMA}} = \mathbf{r}_{\theta} + (t-r) \partial_z \mathbf{u}_{\theta}(\mathbf{u}_{\bar{\theta}}(x_t, t, t) - v(x_t, t)),$$

where $\mathbf{r}_{\theta}$ is the true MeanFlow residual. The FM anchor drives $\mathbf{u}_{\bar{\theta}}(x_t, t, t) \rightarrow v(x_t, t)$, giving $\tilde{\mathbf{r}}^{\text{EMA}} \rightarrow \mathbf{0}$.

Why the target stays unbiased. Propositions 2 and 3 treat the proxy $\hat{v}$ as a tangent-only replacement. A natural alternative is to substitute $\hat{v}$ for the regression target v_{cond} as well, yielding a “fully self-consistent” loss in the spirit of self-distillation. The next proposition shows that the two replacements are not interchangeable: only the tangent admits a gap-attenuated bias.

Proposition 4 (Target/Tangent Bias Asymmetry). *Let $\hat{v} = \mathbf{u}_{\bar{\theta}}(x_t, t, t)$ be a deterministic proxy with bias $\mathbf{b} \triangleq \hat{v} - v(x_t, t)$. Consider two ways of substituting $\hat{v}$ into equation 4:*

(i) Tangent replacement (the $\beta = 1$ corner): *keep the target v_{cond} but replace the JVP tangent. At the loss minimum,*

$$\mathbf{u}^*(x_t, r, t) - v(x_t, t) = -\frac{1}{2}(t-r) \partial_z \mathbf{u}^* \mathbf{b} + \mathcal{O}((t-r) \|\mathbf{b}\|^2),$$

so the stationary error is $\mathcal{O}((t-r) \|\partial_z \mathbf{u}\| \|\mathbf{b}\|)$ and vanishes at $r \rightarrow t$.

(ii) Target replacement: *substitute $\hat{v}$ for v_{cond} in the regression target. At $r = t$ the loss minimum satisfies*

$$\mathbf{u}^*(x_t, t, t) - v(x_t, t) = \mathbf{b},$$

i.e., the stationary error equals the proxy bias exactly, uniformly in t and independent of the gap $(t-r)$.

928 The proof is given in Appendix E. The asymmetry has three immediate consequences. *First*, the
 929 $\beta=1$ corner’s design choice (replace only the tangent, keep $\mathbf{v}_{\text{cond}}$ as target) is not arbitrary: tangent-
 930 replacement bias is multiplicatively damped by the gap $(t-r)$, while target-replacement bias is not.
 931 *Second*, the FM anchor in equation 10 suffices to control the residual tangent-replacement bias
 932 because it pins $\mathbf{u}_\theta(\mathbf{x}_t, t, t) \rightarrow \mathbf{v}$ at the boundary $r=t$, where any target-replacement scheme would
 933 require an analogous anchor at *every* (r, t) . *Third*, the proposition locates a failure mode in self-
 934 distillation-style MeanFlow variants that use a deterministic proxy for both roles: the loss admits
 935 the trivial fixed point $\mathbf{u}^* \equiv \hat{\mathbf{v}}$, which is preserved under proxy updates and only escaped by external
 936 supervision pulling $\hat{\mathbf{v}}$ back toward $\mathbf{v}$ at every t . Concurrent methods that adopt this pattern indeed
 937 report sensitivity to the proxy-update schedule and the boundary regularizer [23, 24].

938 Proposition 2 confirms that the $\mathbf{J}\mathbf{v}'$ amplification of Theorem 2 is eliminated; Proposition 3 confirms
 939 that the residual EMA bias is controlled by the FM anchor, recovering the asymptotic optimum
 940 of Theorem 4; Proposition 4 explains why the FM anchor at the boundary $r=t$ alone is sufficient.

941 E Proofs of Theorems and Propositions

942 E.1 Proof of Theorem 2 (Jacobian Variance Amplification)

943 **Theorem 5** (Jacobian Variance Amplification). *The Jacobi factor amplifies the gradient variance*
 944 *of the original MeanFlow loss:*

$$\text{Var}[\nabla_\theta \ell_{\text{MF}}(\boldsymbol{\theta}) \mid \mathbf{x}_t] \propto \text{Tr}(\mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J}^\top).$$

945 *Proof.* Under the stop-gradient operator, the compound prediction V_θ^{sg} depends on $\boldsymbol{\theta}$ only through
 946 $\mathbf{u}_\theta(\mathbf{x}_t, r, t)$. Let $\nabla_\theta \mathbf{u}_\theta \in \mathbb{R}^{d \times p}$ denote the parameter Jacobian of the network output. The per-step
 947 gradient is

$$\nabla_\theta \ell_{\text{MF}} = 2(\nabla_\theta \mathbf{u}_\theta)^\top (\mathbf{r}_\theta^{\text{sg}} + \mathbf{J}\mathbf{v}'), \quad (12)$$

948 where $\mathbf{r}_\theta^{\text{sg}}$ is the mean-field residual and $\mathbf{J} = (t-r)\partial_z \mathbf{u}_\theta - \mathbf{I}_d$ is the Jacobi factor. All three
 949 quantities $(\nabla_\theta \mathbf{u}_\theta, \mathbf{r}_\theta^{\text{sg}}, \text{ and } \mathbf{J})$ are deterministic conditioned on $\mathbf{x}_t$.

950 Since $\mathbb{E}_{\mathbf{x}_0 \mid \mathbf{x}_t}[\mathbf{v}'] = \mathbf{0}$ by equation 1, the conditional mean gradient is:

$$\mathbb{E}[\nabla_\theta \ell \mid \mathbf{x}_t] = 2(\nabla_\theta \mathbf{u}_\theta)^\top \mathbf{r}_\theta^{\text{sg}}. \quad (13)$$

951 The centered gradient is $\nabla_\theta \ell - \mathbb{E}[\nabla_\theta \ell \mid \mathbf{x}_t] = 2(\nabla_\theta \mathbf{u}_\theta)^\top \mathbf{J}\mathbf{v}'$, so the conditional variance is:

$$\begin{aligned} \text{Var}[\nabla_\theta \ell \mid \mathbf{x}_t] &= \mathbb{E}[\|2(\nabla_\theta \mathbf{u}_\theta)^\top \mathbf{J}\mathbf{v}'\|^2 \mid \mathbf{x}_t] \\ &= 4 \mathbb{E} \left[\left(\mathbf{v}'^\top \mathbf{J}^\top \underbrace{(\nabla_\theta \mathbf{u}_\theta)(\nabla_\theta \mathbf{u}_\theta)^\top}_{\mathbf{G}_\theta \succeq \mathbf{0}} \mathbf{J}\mathbf{v}' \right) \mid \mathbf{x}_t \right] \\ &= 4 \text{Tr}(\mathbf{G}_\theta \mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J}^\top) \propto \text{Tr}(\mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J}^\top), \end{aligned} \quad (14)$$

952 where $\mathbf{G}_\theta = (\nabla_\theta \mathbf{u}_\theta)(\nabla_\theta \mathbf{u}_\theta)^\top \in \mathbb{R}^{d \times d}$ is the parameter-space Gram matrix and the last step uses
 953 $\mathbf{G}_\theta \succeq \mathbf{0}$ to establish proportionality via the cyclic property of the trace. $\square$

954 E.2 Proof of Theorem 3 (Semi-Gradient Gap)

955 **Theorem 6** (Semi-Gradient Gap). *The gradients of the MeanFlow loss with and without the stop-*
 956 *gradient operator differ by*

$$\underbrace{2(t-r) \mathbb{E}_{r,t,\mathbf{x}_0,\mathbf{x}_1} \left[(\nabla_\theta (\partial_z \mathbf{u}_\theta \cdot \mathbf{v}(\mathbf{x}_t, t) + \partial_t \mathbf{u}_\theta))^\top \mathbf{r}_\theta \right]}_{\text{mean-field gradient difference}} + \underbrace{\mathbb{E}_{r,t,\mathbf{x}_0,\mathbf{x}_1} [\nabla_\theta \text{Tr}(\mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J}^\top)]}_{\text{variance-driven gradient difference}}.$$

957 *Proof.* The MeanFlow loss conditioned on $\mathbf{x}_t$ decomposes as (cf. equation 5):

$$\mathbb{E}_{\mathbf{x}_0 \mid \mathbf{x}_t}[\ell_{\text{MF}}] = \|\mathbf{r}_\theta^{\text{sg}}\|^2 + \text{Tr}(\mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J}^\top). \quad (15)$$

958 **Without stop-gradient.** Both terms depend on θ (through $\mathbf{u}_\theta$ and $\partial_z \mathbf{u}_\theta$), so the full gradient is:

$$\nabla_\theta \mathbb{E}[\ell] = \nabla_\theta \|\mathbf{r}_\theta\|^2 + \nabla_\theta \text{Tr}(\mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J}^\top). \quad (16)$$

959 **With stop-gradient.** The JVP terms in $\mathbf{r}_\theta^{\text{sg}}$ and $\mathbf{J}$ are treated as constants, so:

$$\nabla_\theta^{\text{sg}} \|\mathbf{r}^{\text{sg}}\|^2 = 2 (\nabla_\theta \mathbf{u}_\theta)^\top \mathbf{r}^{\text{sg}}, \quad \nabla_\theta^{\text{sg}} \text{Tr}(\mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J}^\top) = \mathbf{0}. \quad (17)$$

960 The gap between the full and stop-gradient gradients is therefore:

$$\begin{aligned} & \nabla_\theta \mathbb{E}[\ell] - \nabla_\theta^{\text{sg}} \mathbb{E}[\ell] \\ &= \underbrace{\nabla_\theta \|\mathbf{r}_\theta\|^2 - 2 (\nabla_\theta \mathbf{u}_\theta)^\top \mathbf{r}^{\text{sg}}}_{\text{mean-field gradient difference}} + \underbrace{\nabla_\theta \text{Tr}(\mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J}^\top)}_{\text{variance-driven difference}}. \end{aligned} \quad (18)$$

961 Expanding the first term: $\nabla_\theta \|\mathbf{r}\|^2 = 2(\nabla_\theta \mathbf{r}_\theta)^\top \mathbf{r}_\theta$, and $\nabla_\theta \mathbf{r}_\theta$ includes the derivative of the JVP
962 terms $(t-r)(\partial_z \mathbf{u}_\theta \cdot \mathbf{v} + \partial_t \mathbf{u}_\theta)$ w.r.t. θ , yielding the second-order mean-field difference $2(t-r)$
963 $\mathbb{E}[(\nabla_\theta(\partial_z \mathbf{u}_\theta \cdot \mathbf{v} + \partial_t \mathbf{u}_\theta))^\top \mathbf{r}_\theta]$. At convergence $\mathbf{r}_\theta \rightarrow \mathbf{0}$, this term vanishes, and the gap is
964 dominated by the variance-driven term $\nabla_\theta \text{Tr}(\mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J}^\top)$. $\square$

965 E.3 Proof of Proposition 2 (Gradient under the $\beta=1$ corner loss)

966 **Proposition 7** (Gradient under the $\beta=1$ corner loss). *The gradient of $\mathcal{L}_{\text{MF}}^{\text{EMA}}$ takes the form*

$$\nabla_\theta \mathcal{L}_{\text{MF}}^{\text{EMA}} = 2 \mathbb{E}[(\nabla_\theta \mathbf{u}_\theta(\mathbf{x}_t, r, t))^\top \tilde{\mathbf{r}}^{\text{EMA}}],$$

967 where $\tilde{\mathbf{r}}^{\text{EMA}} := V_\theta^{\text{EMA}} - \mathbf{v}(\mathbf{x}_t, t)$ is the mean-field residual under the EMA tangent, and no $\mathbf{J}\mathbf{v}'$ noise
968 term appears.

969 *Proof.* The EMA velocity anchor $\mathbf{u}_\theta(\mathbf{x}_t, t, t)$ is deterministic given $\mathbf{x}_t$, since the EMA paramete-
970 rs θ are fixed during the gradient step. Under stop-gradient, V_θ^{EMA} depends on θ only through
971 $\mathbf{u}_\theta(\mathbf{x}_t, r, t)$. By the same argument as equation 12, the per-step gradient is:

$$\nabla_\theta \ell = 2 (\nabla_\theta \mathbf{u}_\theta)^\top (V_\theta^{\text{EMA}} - \mathbf{v}_{\text{cond}}). \quad (19)$$

972 Writing $\mathbf{v}_{\text{cond}} = \mathbf{v}(\mathbf{x}_t, t) + \mathbf{v}'$ and noting that $\tilde{\mathbf{r}}^{\text{EMA}} := V_\theta^{\text{EMA}} - \mathbf{v}(\mathbf{x}_t, t)$ is deterministic given $\mathbf{x}_t$:

$$\mathbb{E}_{\mathbf{x}_0|\mathbf{x}_t}[\nabla_\theta \ell] = 2 (\nabla_\theta \mathbf{u}_\theta)^\top \tilde{\mathbf{r}}^{\text{EMA}}, \quad (20)$$

973 since $\mathbb{E}[\mathbf{v}' | \mathbf{x}_t] = \mathbf{0}$. The tower property $\nabla_\theta \mathcal{L}_{\text{MF}}^{\text{EMA}} = \mathbb{E}_{r,t,\mathbf{x}_t}[\mathbb{E}_{\mathbf{x}_0|\mathbf{x}_t}[\nabla_\theta \ell]]$ gives the stated result.
974 Crucially, no $\mathbf{J}\mathbf{v}'$ term appears because the EMA tangent is deterministic; compare with the original
975 MeanFlow gradient equation 12 where the stochastic tangent produces the $\mathbf{J}\mathbf{v}'$ noise. $\square$

976 E.4 Proof of Proposition 3 (Bias-Variance Tradeoff)

977 **Proposition 8** (Bias-Variance Tradeoff). *The EMA mean-field residual decomposes as*

$$\tilde{\mathbf{r}}^{\text{EMA}} = \mathbf{r}_\theta + (t-r) \partial_z \mathbf{u}_\theta(\mathbf{u}_\theta(\mathbf{x}_t, t, t) - \mathbf{v}(\mathbf{x}_t, t)),$$

978 where $\mathbf{r}_\theta$ is the true MeanFlow residual from equation 5. At convergence $\mathbf{r}_\theta \rightarrow \mathbf{0}$ and the FM anchor
979 drives $\mathbf{u}_\theta(\mathbf{x}_t, t, t) \rightarrow \mathbf{v}(\mathbf{x}_t, t)$, giving $\tilde{\mathbf{r}}^{\text{EMA}} \rightarrow \mathbf{0}$.

980 *Proof.* Expanding V_θ^{EMA} from equation 9, the value of the compound prediction (ignoring the stop-
981 gradient, which does not affect values) is:

$$V_\theta^{\text{EMA}} = \mathbf{u}_\theta + (t-r)(\partial_z \mathbf{u}_\theta \cdot \mathbf{u}_\theta(\mathbf{x}_t, t, t) + \partial_t \mathbf{u}_\theta). \quad (21)$$

982 The true MeanFlow residual (with the marginal velocity as tangent) is:

$$\mathbf{r}_\theta = \mathbf{u}_\theta + (t-r)(\partial_z \mathbf{u}_\theta \cdot \mathbf{v}(\mathbf{x}_t, t) + \partial_t \mathbf{u}_\theta) - \mathbf{v}(\mathbf{x}_t, t). \quad (22)$$

983 Subtracting:

$$\begin{aligned} \tilde{\mathbf{r}}^{\text{EMA}} &= V_\theta^{\text{EMA}} - \mathbf{v}(\mathbf{x}_t, t) \\ &= \mathbf{r}_\theta + (t-r) \partial_z \mathbf{u}_\theta(\mathbf{u}_\theta(\mathbf{x}_t, t, t) - \mathbf{v}(\mathbf{x}_t, t)). \end{aligned} \quad (23)$$

984 At convergence, the true residual $\mathbf{r}_\theta \rightarrow \mathbf{0}$ (the MeanFlow identity is satisfied) and the FM anchor
985 loss equation 10 supervises the boundary condition, driving $\mathbf{u}_\theta(\mathbf{x}_t, t, t) \rightarrow \mathbf{v}(\mathbf{x}_t, t)$ and hence
986 $\tilde{\mathbf{r}}^{\text{EMA}} \rightarrow \mathbf{0}$. $\square$

987 E.5 Proof of Proposition 4 (Target/Tangent Bias Asymmetry)

988 **Proposition 9** (Target/Tangent Bias Asymmetry). *Let $\hat{\mathbf{v}} = \mathbf{u}_{\hat{\theta}}(\mathbf{x}_t, t, t)$ be a deterministic proxy with*
 989 *bias $\mathbf{b} \triangleq \hat{\mathbf{v}} - \mathbf{v}(\mathbf{x}_t, t)$. For the tangent-replacement loss (the $\beta = 1$ corner), the loss minimum $\mathbf{u}^*$*
 990 *satisfies $\mathbf{u}^*(\mathbf{x}_t, r, t) - \mathbf{v}(\mathbf{x}_t, t) = -\frac{1}{2}(t-r) \partial_{\mathbf{z}} \mathbf{u}^* \mathbf{b} + \mathcal{O}((t-r) \|\mathbf{b}\|^2)$. For the target-replacement*
 991 *loss, the loss minimum at $r=t$ satisfies $\mathbf{u}^*(\mathbf{x}_t, t, t) - \mathbf{v}(\mathbf{x}_t, t) = \mathbf{b}$.*

992 *Proof. Tangent replacement.* Substituting $\hat{\mathbf{v}}$ for $\mathbf{v}_{\text{cond}}$ only in the JVP tangent (target unchanged)
 993 gives the per-sample compound prediction

$$V_{\theta}^{\text{EMA}} = \mathbf{u}_{\theta} + (t-r)(\partial_{\mathbf{z}} \mathbf{u}_{\theta} \cdot \hat{\mathbf{v}} + \partial_t \mathbf{u}_{\theta}),$$

994 and the loss is $\mathbb{E} \|V_{\theta}^{\text{EMA}} - \mathbf{v}_{\text{cond}}\|^2$. By stationarity, $\mathbb{E}[V_{\theta}^{\text{EMA}} - \mathbf{v}_{\text{cond}} \mid \mathbf{x}_t] = \mathbf{0}$ at the minimum. Taking
 995 conditional expectation and using $\mathbb{E}[\mathbf{v}_{\text{cond}} \mid \mathbf{x}_t] = \mathbf{v}(\mathbf{x}_t, t)$,

$$\mathbf{u}^* + (t-r)(\partial_{\mathbf{z}} \mathbf{u}^* \cdot \hat{\mathbf{v}} + \partial_t \mathbf{u}^*) = \mathbf{v}(\mathbf{x}_t, t).$$

996 At the same point, the *true* MeanFlow identity (with the marginal velocity as tangent) reads

$$\mathbf{u}^{\diamond} + (t-r)(\partial_{\mathbf{z}} \mathbf{u}^{\diamond} \cdot \mathbf{v} + \partial_t \mathbf{u}^{\diamond}) = \mathbf{v}(\mathbf{x}_t, t),$$

997 with $\mathbf{u}^{\diamond}(\mathbf{x}_t, t, t) = \mathbf{v}(\mathbf{x}_t, t)$. Subtracting and writing $\mathbf{u}^* = \mathbf{u}^{\diamond} + \Delta$,

$$\Delta + (t-r) \partial_{\mathbf{z}} \mathbf{u}^* \mathbf{b} + (t-r) D_t \Delta = \mathbf{0}, \quad D_t \Delta \triangleq \partial_t \Delta + \mathbf{v} \cdot \partial_{\mathbf{z}} \Delta,$$

998 with the boundary condition $\Delta(\mathbf{x}_t, t, t) = \mathbf{0}$. Substituting $s = t-r$ at fixed r converts this to a
 999 first-order linear ODE in s :

$$s \frac{d\Delta}{ds} + \Delta = -s \partial_{\mathbf{z}} \mathbf{u}^* \mathbf{b} + \mathcal{O}(\|\mathbf{b}\|^2),$$

1000 which has $s\Delta(s) = -\frac{s^2}{2} \partial_{\mathbf{z}} \mathbf{u}^* \mathbf{b} + C$; the boundary condition $\Delta(0) = \mathbf{0}$ forces $C = 0$. Hence

$$\Delta(\mathbf{x}_t, r, t) = -\frac{1}{2}(t-r) \partial_{\mathbf{z}} \mathbf{u}^* \mathbf{b} + \mathcal{O}((t-r) \|\mathbf{b}\|^2),$$

1001 i.e., the stationary error is $\mathcal{O}((t-r) \|\partial_{\mathbf{z}} \mathbf{u}^*\| \|\mathbf{b}\|)$ and vanishes at the boundary $r \rightarrow t$. The factor of $\frac{1}{2}$
 1002 comes from integrating the material-derivative term, which contributes equally to the leading-order
 1003 coefficient as the bias source term.

1004 **Target replacement.** Substituting $\hat{\mathbf{v}}$ for $\mathbf{v}_{\text{cond}}$ in the regression target yields the loss $\mathbb{E} \|\mathbf{u}_{\theta} + (t-r)$
 1005 $r) D\mathbf{u}/Dt - \hat{\mathbf{v}}\|^2$. By stationarity, the loss minimum satisfies $\mathbb{E}[\mathbf{u}^* + (t-r) D\mathbf{u}^*/Dt - \hat{\mathbf{v}} \mid \mathbf{x}_t] = \mathbf{0}$.
 1006 Specializing to $r=t$, the gap term vanishes and the equation collapses to $\mathbf{u}^*(\mathbf{x}_t, t, t) = \hat{\mathbf{v}} = \mathbf{v}(\mathbf{x}_t, t) +$
 1007 $\mathbf{b}$, so the stationary error at the boundary is exactly $\mathbf{b}$, independent of (t, r) .

1008 **Asymmetry.** Tangent replacement carries a multiplicative $(t-r)$ factor on the bias and is therefore
 1009 pinned to zero at the boundary $r=t$ that the FM anchor enforces; target replacement carries no such
 1010 factor and the bias persists at every t , requiring a boundary regularizer at *every* (r, t) to remove. $\square$

1011 E.6 Proof of Theorem 4 (Optimal Control-Variate Coefficient)

1012 **Theorem 10** (Optimal control-variate coefficient). *Under the scalar-isotropic approximation $\Sigma_{\mathbf{v}'} \approx$*
 1013 *$\sigma^2 \mathbf{I}_d$ and $\mathbf{J} \approx \kappa \mathbf{I}_d$, the conditional MSE of $\mathbf{g}^{(\beta)}$ in equation 7 is*

$$M(\beta) = \beta^2 (\kappa+1)^2 \|\mathbf{b}\|^2 + \sigma^2 d ((1-\beta)\kappa - \beta)^2,$$

1014 *with unique minimizer $\beta^* = \frac{\kappa}{\kappa+1} \cdot \frac{\sigma^2 d}{\sigma^2 d + \|\mathbf{b}\|^2}$ and optimum value $M(\beta^*) = \sigma^2 d \kappa^2 \|\mathbf{b}\|^2 / (\sigma^2 d + \|\mathbf{b}\|^2)$.*
 1015 *For all κ , $\|\mathbf{b}\|^2 > 0$, the optimum strictly dominates both $M(0) = \sigma^2 \kappa^2 d$ and $M(1) = (\kappa+1)^2 \|\mathbf{b}\|^2 +$*
 1016 *$\sigma^2 d$.*

1017 *Proof. Step 1: bias-variance decomposition.* From equation 7,

$$\mathbf{g}^{(\beta)} = 2 \mathbf{g}^{\top} \left[\bar{\mathbf{r}} + ((1-\beta) \mathbf{J} - \beta \mathbf{I}) \mathbf{v}' + \beta (\mathbf{J} + \mathbf{I}) \mathbf{b} \right].$$

Using $\mathbb{E}_{\mathbf{x}_0|\mathbf{x}_t}[\mathbf{v}'] = \mathbf{0}$, the conditional mean is $\mathbb{E}[\mathbf{g}^{(\beta)} | \mathbf{x}_t] = 2\mathbf{g}^\top(\bar{\mathbf{r}} + \beta(\mathbf{J} + \mathbf{I})\mathbf{b})$ and the centered noise is $-2\mathbf{g}^\top((1-\beta)\mathbf{J} - \beta\mathbf{I})\mathbf{v}'$ (relative to the ideal target $2\mathbf{g}^\top\bar{\mathbf{r}}$). The conditional MSE w.r.t. the ideal target therefore decomposes as

$$\text{MSE} = \underbrace{4\beta^2 \|\mathbf{g}^\top(\mathbf{J} + \mathbf{I})\mathbf{b}\|^2}_{\text{bias}^2} + \underbrace{4 \text{Tr}(\mathbf{g}\mathbf{g}^\top((1-\beta)\mathbf{J} - \beta\mathbf{I})\Sigma_{\mathbf{v}'}((1-\beta)\mathbf{J} - \beta\mathbf{I})^\top)}_{\text{variance}}.$$

Step 2: matrix-form expansion and strict convexity. Let $\mathbf{A} \triangleq \mathbf{J} + \mathbf{I}$ and $\mathbf{G}_\theta \triangleq \mathbf{g}\mathbf{g}^\top \succeq \mathbf{0}$. Substituting $((1-\beta)\mathbf{J} - \beta\mathbf{I}) = \mathbf{J} - \beta\mathbf{A}$ and using $\|\mathbf{g}^\top\mathbf{A}\mathbf{b}\|^2 = \text{Tr}(\mathbf{G}_\theta\mathbf{A}\mathbf{b}\mathbf{b}^\top\mathbf{A}^\top)$, the conditional MSE writes as a quadratic in β :

$$M(\beta) = \alpha_0 - 2\alpha_1\beta + \alpha_2\beta^2, \quad (24)$$

with (suppressing the constant prefactor 4)

$$\begin{aligned} \alpha_0 &\triangleq \text{Tr}(\mathbf{G}_\theta \mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J}^\top), \\ \alpha_1 &\triangleq \text{Tr}(\mathbf{G}_\theta \mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{A}^\top), \\ \alpha_2 &\triangleq \text{Tr}(\mathbf{G}_\theta \mathbf{A} (\Sigma_{\mathbf{v}'} + \mathbf{b}\mathbf{b}^\top) \mathbf{A}^\top). \end{aligned}$$

(We used the symmetry of $\mathbf{G}_\theta$ and $\Sigma_{\mathbf{v}'}$ to consolidate the linear-in- β trace into a single α_1 .) The Hessian is

$$\frac{\partial^2 M}{\partial \beta^2} = 2\alpha_2 = 2 \text{Tr}(\mathbf{G}_\theta \mathbf{A} (\Sigma_{\mathbf{v}'} + \mathbf{b}\mathbf{b}^\top) \mathbf{A}^\top) \geq 0,$$

since each factor inside the trace is positive semi-definite ($\mathbf{G}_\theta \succeq \mathbf{0}$, $\Sigma_{\mathbf{v}'} + \mathbf{b}\mathbf{b}^\top \succeq \mathbf{0}$, and $\mathbf{A}(\Sigma_{\mathbf{v}'} + \mathbf{b}\mathbf{b}^\top)\mathbf{A}^\top = \mathbf{M}\mathbf{M}^\top \succeq \mathbf{0}$ for $\mathbf{M} \triangleq \mathbf{A}(\Sigma_{\mathbf{v}'} + \mathbf{b}\mathbf{b}^\top)^{1/2}$). Hence $M(\beta)$ is convex in β . The Hessian is strictly positive whenever $\mathbf{G}_\theta^{1/2}\mathbf{M} \neq \mathbf{0}$, which holds outside two pathological regimes: (i) $\mathbf{A} = \mathbf{0}$, equivalently $\mathbf{J} = -\mathbf{I}$ (network with $\partial_{\mathbf{z}}\mathbf{u}_\theta \equiv \mathbf{0}$, e.g., at saturated initialization), in which case the linear coefficient α_1 also vanishes and $M(\beta)$ is constant; (ii) $\mathbf{g} = \mathbf{0}$ (frozen parameters). Outside these, $M(\beta)$ is *strictly* convex with unique unconstrained minimizer

$$\beta_{\text{matrix}}^* = \frac{\alpha_1}{\alpha_2} = \frac{\text{Tr}(\mathbf{G}_\theta \mathbf{J} \Sigma_{\mathbf{v}'} (\mathbf{J} + \mathbf{I})^\top)}{\text{Tr}(\mathbf{G}_\theta (\mathbf{J} + \mathbf{I}) (\Sigma_{\mathbf{v}'} + \mathbf{b}\mathbf{b}^\top) (\mathbf{J} + \mathbf{I})^\top)}. \quad (25)$$

Note that β_{matrix}^* may fall outside $[0, 1]$; the box-constrained optimum on the $\beta \in [0, 1]$ interval (cf. equation 6) is then $\text{clip}(\beta_{\text{matrix}}^*, 0, 1)$, attained at the nearer endpoint by convexity.

Step 3: scalar approximation. Substituting $\mathbf{J} = \kappa\mathbf{I}_d$, $\Sigma_{\mathbf{v}'} = \sigma^2\mathbf{I}_d$, $\mathbf{A} = (\kappa+1)\mathbf{I}_d$, and the parameter-isotropy approximation $\mathbf{G}_\theta \propto \mathbf{I}_d$ (which absorbs the parameter-direction normalization into a per-component MSE),

$$\alpha_0 \propto \sigma^2\kappa^2d, \quad \alpha_1 \propto \sigma^2\kappa(\kappa+1)d, \quad \alpha_2 \propto (\kappa+1)^2(\sigma^2d + \|\mathbf{b}\|^2),$$

which equivalently gives

$$M(\beta) \propto \beta^2(\kappa+1)^2\|\mathbf{b}\|^2 + \sigma^2d((1-\beta)\kappa - \beta)^2.$$

Step 4: minimization. Differentiating and setting to zero (or directly evaluating α_1/α_2),

$$\beta^* = \frac{\alpha_1}{\alpha_2} = \frac{\sigma^2\kappa(\kappa+1)d}{(\kappa+1)^2(\sigma^2d + \|\mathbf{b}\|^2)} = \frac{\kappa}{\kappa+1} \cdot \frac{\sigma^2d}{\sigma^2d + \|\mathbf{b}\|^2}.$$

Step 5: optimum value and corner dominance. At β^* , $(1-\beta^*)\kappa - \beta^* = \kappa\|\mathbf{b}\|^2/(\sigma^2d + \|\mathbf{b}\|^2)$. Substituting back,

$$M(\beta^*) \propto \frac{\sigma^2d\kappa^2\|\mathbf{b}\|^2}{\sigma^2d + \|\mathbf{b}\|^2}.$$

For $\kappa > 0$ and $\|\mathbf{b}\|^2 > 0$: $M(\beta^*)/M(0) = \|\mathbf{b}\|^2/(\sigma^2d + \|\mathbf{b}\|^2) < 1$, so $M(\beta^*) < M(0)$. For the $M(1)$ comparison, $M(1) - M(\beta^*) \propto (\kappa+1)^2\|\mathbf{b}\|^2 + \sigma^2d - \sigma^2d\kappa^2\|\mathbf{b}\|^2/(\sigma^2d + \|\mathbf{b}\|^2)$, and the first two terms exceed $\sigma^2d\kappa^2\|\mathbf{b}\|^2/(\sigma^2d + \|\mathbf{b}\|^2)$ since $(\kappa+1)^2 > \kappa^2$ for $\kappa > 0$. Strict dominance is therefore over the unconstrained interior optimum; if $\beta_{\text{matrix}}^* \notin [0, 1]$ the box-constrained optimum is at the nearer corner and the dominance is non-strict. $\square$

F DiT Latent-Space Training on ImageNet

We test whether the variance-reduction mechanism of Theorems 2 and 4 scales beyond toys by training DiT [42] variants on ImageNet in the latent space of a frozen Stable Diffusion VAE, following the recipe of [21]. The four β configurations share identical hyperparameters and code paths; only the loss differs (vanilla MeanFlow MSE for the interior β runs versus the EMA-tangent loss in equation 11 for the $\beta = 1$ corner).

F.1 DiT-B/4, ImageNet-256 $\times$ 256

Setup: logit-normal (r, t) sampling with mean -0.4 and stddev 1 , overlap rate 0.75 , AdamW with constant learning rate 10^{-4} , EMA decay 0.9999 , batch size 256 across 32 TPU v4 chips. All four configurations ($\beta \in \{0, 0.25, 0.5, 1\}$) share the same DiT-B/4 backbone, data pipeline, optimizer, EMA schedule, and (r, t) sampling distribution. The interior β runs ($\beta \in \{0, 0.25, 0.5\}$) reuse the baseline configuration verbatim and add only the tangent-mixing rule of equation 6; the $\beta = 1$ corner additionally disables the Karras-style adaptive loss weighting and adds the FM anchor of equation 10 ($\delta \sim \mathcal{U}[0, 10^{-3}]$, $\lambda = 0.1$). The extra EMA forward pass and FM-anchor evaluation per step add up to a $\sim 22\%$ wall-clock overhead on TPU v4 (2.20 steps/s for $\beta = 1$ versus 2.65 steps/s for the baseline); the $\beta \in \{0.25, 0.5\}$ interior runs incur only the EMA forward overhead ($\sim 10\%$).

Three-floor structure and matched-step ordering. Table 5 reports FID_{50k} at representative 5k-step checkpoints for all three runs, and Figure 5 renders the full trajectories on a semilog-y axis. After the early-training transient ($\leq 10k$ steps), the empirical ordering $\text{FID}(\beta=0) < \text{FID}(\beta=0.5) < \text{FID}(\beta=1)$ holds at *every* matched-step checkpoint we logged (24 consecutive datapoints from step 30k onward). The trajectory has two phases: in the first (steps 20k to 45k), the gap between $\beta=0.5$ and the $\beta=1$ corner widens monotonically (from $+2.1$ at step 20k to -11.4 at step 45k), validating the interior optimum predicted by Theorem 4; once the gradient-noise reduction of $\beta=0.5$ has paid off, the convergence-rate advantage stabilizes. In the second phase ($\geq 90k$), the gap between $\beta=0.5$ and the $\beta=0$ baseline reaches an asymptotic offset of $+7.77 \pm 0.11$ FID over 11 consecutive matched-step windows (steps 95k to 145k), well below the $+16$ FID gap of the early-training phase.

$\beta = 0.5$ beats $\beta = 1$'s asymptotic floor. The $\beta = 1$ corner run completed its scheduled 300k-step training with a final logged FID of **23.36** (step 295k); we treat this as the empirical $\beta = 1$ floor. The $\beta = 0.5$ run also completed its scheduled 300k steps with a final logged FID of **12.51** (step 295k), a margin of 10.85 FID below the $\beta = 1$ asymptote. The four converged FID floors recover the bias-variance ordering of Theorem 4:

$$\text{FID}^*(\beta=0) \approx 10.34 < \text{FID}^*(\beta=0.25) \lesssim 14.79 < \text{FID}^*(\beta=0.5) = 12.51 < \text{FID}^*(\beta=1) = 23.36.$$

(The baseline minimum FID of 10.34 was logged at step 145k; subsequent training showed mild late-stage drift to 12.12 at step 235k, so the asymptotic baseline floor lies in $[10.3, 12.3]$.) The $\beta=0.25$ run is still in training; its latest logged FID at step 170k is 14.79 and remains a strict upper bound on its own asymptote.

Quantitatively, taking offsets from the best-logged baseline floor 10.34, $\Delta\text{FID}(\beta=0.5) \approx 2.17$ and $\Delta\text{FID}(\beta=1) \approx 13.02$, giving an empirical FID-offset ratio of $\sim 6:1$. The gradient-MSE bias term $\beta^2(\kappa+1)^2\|\mathbf{b}\|^2$ predicts the much smaller 1:4 ratio (from β^2 alone). The FID landscape is therefore *super-linear* in the MSE-axis bias: it penalizes large bias more aggressively than gradient MSE does, which pulls the FID-optimal β past the gradient-MSE interior minimizer to the unbiased corner $\beta=0$. The four-floor structure is the strongest empirical confirmation of the bias-variance decomposition we obtain at DiT scale.

Bias and shrinkage trajectory. To localize the predicted optimum we measure two diagnostics on the DiT checkpoints: an EMA-tracking proxy $\|\widehat{\mathbf{b}}\|^2 \triangleq \mathbb{E}_{\mathbf{x}_t} [\|\mathbf{u}_\theta(\mathbf{x}_t, t, t) - \mathbf{u}_{\bar{\theta}}(\mathbf{x}_t, t, t)\|^2]$ (the boundary gap between current parameters and the EMA copy), and the irreducible noise floor $\sigma^2 d \approx 8.2 \times 10^3$ from $\text{Tr}(\Sigma_{v'})$ on the same batch. Table 6 reports the trajectory: the EMA-tracking proxy *decays* through training, and the shrinkage factor $\sigma^2 d / (\sigma^2 d + \|\widehat{\mathbf{b}}\|^2)$ stays near 1 throughout converged training. We caveat that $\|\widehat{\mathbf{b}}\|^2$ is the EMA-tracking gap, not the true model bias

Table 5: DiT-B/4 / ImageNet-256 FID_{50k} (lower is better) at representative 5k-step checkpoints across the four-point β -sweep. The baseline ($\beta=0$) and $\beta=1$ corner runs were trained to completion (235k and 295k logged steps, respectively); the interior probe at $\beta=0.5$ completed its scheduled 300k steps with final FID 12.51 at step 295k; the interior probe at $\beta=0.25$ is still training at the time of submission, with the latest logged FID at step 145k. Bottom rows give the matched-step deltas. The four-point ordering $\text{FID}(\beta=0) < \text{FID}(\beta=0.25) < \text{FID}(\beta=0.5) < \text{FID}(\beta=1)$ holds at every matched-step checkpoint.

step (k)	30	40	50	60	80	100	115	130	145	final
baseline ($\beta=0$)	128.9	79.2	54.5	41.8	28.3	17.7	14.4	12.1	10.3	12.1 [†]
$\beta=0.25$	138.5	88.4	61.6	46.7	30.4	23.3	20.3	18.3	16.7	(running) [‡]
$\beta=0.5$	145.2	94.6	n/a	n/a	n/a	25.5	22.2	19.7	18.1	12.5
$\beta=1$	148.8	104.1	79.9	65.7	49.9	41.7	38.0	35.2	32.7	23.4
$\Delta(\beta=0.25 - \text{baseline})$	+9.6	+9.2	+7.1	+4.9	+2.1	+5.6	+5.9	+6.2	+6.4	n/a
$\Delta(\beta=0.5 - \text{baseline})$	+16.3	+15.4	n/a	n/a	n/a	+7.8	+7.8	+7.6	+7.8	n/a
$\Delta(\beta=1 - \text{baseline})$	+19.9	+24.9	+25.4	+23.9	+21.6	+24.0	+23.6	+23.1	+22.4	n/a

[†] The baseline run was stopped at step 235k with last logged FID 12.12; its minimum logged FID was 10.34 at step 145k, with mild late-training drift thereafter. We report the late-training value as the empirical asymptotic floor for the matched-step comparison.

[‡] Latest logged $\beta=0.25$ FID at the time of submission is 14.79 at step 170k, still descending; the run is scheduled for 300k total steps.

1095 $\|\mathbf{u}_\theta(\mathbf{x}_t, t, t) - \mathbf{v}(\mathbf{x}_t, t)\|^2$; cleanly estimating the latter requires access to the conditional distribu-
1096 tion $p(\mathbf{x}_0 | \mathbf{x}_t)$, which is intractable on ImageNet. Combined with $\kappa \approx 1$ from the Frobenius-norm
1097 Hutchinson trace, the scalar-isotropic substitution gives $\beta^* \in [0.46, 0.50]$ across the full trajectory;
1098 this is a coarse first-cut prediction that the direct matrix-form measurement (Section 5.4) refines to
1099 $\beta_{\text{no bias}}^* \approx 0.94$.

Table 6: Direct DiT-checkpoint measurement of the bias-variance decomposition ingredients of Theorem 4 at $t = 0.5$. The EMA bias $\|\mathbf{b}\|^2$ averages over 128 synthetic-Gaussian $\mathbf{x}_t$ samples and is reported for both methods (which track within 20% of each other). The shrinkage factor uses $\sigma^2 d \approx 8.2 \times 10^3$. The predicted β^* uses $\kappa = 1$ from the Hutchinson trace.

step (k)	$\ \mathbf{b}\ ^2$ (avg)	$\sigma^2 d / (\sigma^2 d + \ \mathbf{b}\ ^2)$	β^*
20	~ 700	0.92	0.46
40	~ 150	0.98	0.49
80	~ 50	0.99	0.50